mathematics

IMPACT FACTOR 2.3

CITESCORE 5.4

Article

Hypercomplex Numbers—A Tool for Enhanced Efficiency and Intelligence in Digital Signal Processing

Zlatka Valkova-Jarvis, Maria Nenova and Dimitriya Mihaylova

Special Issue
Computational Intelligence in Communication Networks

Edited by
Prof. Dr. Georgi Iliev

MDPI

https://doi.org/10.3390/math13030504

mathematics

Article

Hypercomplex Numbers—A Tool for Enhanced Efficiency and Intelligence in Digital Signal Processing

Zlatka Valkova-Jarvis *, Maria Nenova and Dimitriya Mihaylova

Faculty of Telecommunications, Technical University of Sofia, 1000 Sofia, Bulgaria; mvn@tu-sofia.bg (M.N.); dam@tu-sofia.bg (D.M.)
* Correspondence: zvv@tu-sofia.bg

Abstract: Mathematics is the wide-ranging solid foundation of the engineering sciences which ensures their progress by providing them with its unique toolkit of rules, methods, algorithms and numerical systems. In this paper, an overview of the numerical systems that have currently found an application in engineering science and practice is offered, while also mentioning those systems that still await full and comprehensive applicability, recognition, and acknowledgment. Two possible approaches for representing hypercomplex numbers are proposed—based on real numbers and based on complex numbers. This makes it possible to justify the creation and introduction of numerical systems specifically suited to digital signal processing (DSP), which is the basis of all modern technical sciences ensuring the technological progress of mankind. Understanding the specifics, peculiarities, and properties of the large and diverse family of hypercomplex numbers is the first step towards their more comprehensive and thorough study, and hence their use in a number of high-tech intelligent applications in various engineering and scientific fields, such as information and communication technologies (ICT), communication and neural networks, cybersecurity and national security, artificial intelligence (AI), space and military technologies, industrial engineering and machine learning, astronomy, applied mathematics, quantum physics, etc. The issues discussed in this paper are, however, far from exhausting the scientific topics related to both hypercomplex numbers in general and those relevant to DSP. This is a promising scientific area, the potential of which has not yet been fully explored, but research already shows the enhanced computational efficiency and intelligent performance of hypercomplex DSP.

Keywords: real; complex; hypercomplex numbers; quaternions; biquaternions; bicomplex numbers; digital signal processing; Cayley–Dickson algebra

MSC: 11R52; 62P30

Academic Editor: Jonathan Blackledge

Received: 8 January 2025
Revised: 24 January 2025
Accepted: 29 January 2025
Published: 3 February 2025

Citation: Valkova-Jarvis, Z.; Nenova, M.; Mihaylova, D. Hypercomplex Numbers—A Tool for Enhanced Efficiency and Intelligence in Digital Signal Processing. *Mathematics* **2025**, *13*, 504. https://doi.org/10.3390/math13030504

1. Introduction

Generally speaking, the science of mathematics is at least one developmental step ahead in comparison to the technical sciences, and this is a necessity, given that it is mathematics which is the main theoretical basis for these sciences. Numerical systems are no exception to this rule and can be classified as follows: real, complex, and hypercomplex numbers.

Hypercomplex numbers are a natural mathematical extension of complex numbers. As with most new mathematical theories, hypercomplex numbers did not find a practical application immediately after their discovery. A seminal moment that transformed them from a pure mathematical concept into a theoretical basis of engineering research was, for example, the notion of them being able to represent 3D rotation, leading to the earliest applications of hypercomplex numbers in fields such as computer graphics, coordinated 3D image processing, aerospace engineering for stabilization, and altitude control. Currently, the applications of this intricate but very powerful mathematical tool are numerous and far reaching, and include the following: robotics, aerospace, telemedicine, air and sea navigation, seismology, meteorology, microbiology, geology, etc. One of the most intensively developing application areas of hypercomplex numbers is telecommunications. Over the last thirty years, there has been an increased interest in this number system, which is proving suitable for the development and improvement of many of today's telecommunication technologies. Numerous research papers have reported on problems related to machine-to-machine (M2M) communications, Low-Power Wide-Area Networks (LPWANs), satellite communications, facial recognition, neural networks, medical imaging, moving object data processing, color image analysis, segmentation and restoration, color watermarking, remote speech recognition, 3D audio systems, the localization and reconstruction of 3D objects, noise estimation in color images, digital wavelet processing, radar and sonar signals, encryption systems, filter banks for the parallel processing of 2D and 3D signals, beamforming, interference and interference protection, adaptive filtering of 3D and 4D signals, intelligent Orthogonal Frequency-Division Multiplexing (OFDM) telecommunication systems, and many more. These applications have something in common—for all of them, DSP is a fundamental theoretical basis and tool.

The use of hypercomplex numbers in the field of DSP began at the dawn of the development of this science around the middle of the 20th century. As an example, research based on quaternions and Hamiltonian biquaternions has brought to the foreground some of their shortcomings and shown that they are not always suitable for DSP applications. This has led to the development and introduction of various new hypercomplex numerical systems, such as the so-called reduced biquaternions. These, in essence, are bicomplex numbers possessing the commutativity property, which is of prime importance for DSP algorithms. Algorithms with bicomplex coefficients, often referred to by the generic name hypercomplex, have become particularly popular in recent years because of their advantages in processing signals represented by real, complex, or hypercomplex numbers. It is well recognised that complex DSP algorithms demonstrate greater efficiency, intelligence, and smartness compared to real DSP algorithms. Similarly, hypercomplex DSP could bring additional computational efficiency to the methods, increased intelligence in processing, and could possibly further facilitate their implementation.

Numerical systems and their applications have been discussed in many publications, some of which will be mentioned in this work. However, it would be useful to provide a summary of these issues, and that is the aim of this paper. It approaches the topic by first reviewing the number systems in detail and their representation in general. Then, the subset of numbers applicable to DSP is defined, clarifying their place among the large family of hypercomplex numbers and the properties they require in order to be used in DSP.

This paper is organized in the following way: numerical systems based on real numbers are introduced in Section 2. Section 3 outlines the essence and main properties of numerical systems based on complex numbers. Numerical systems applicable to DSP are presented in Section 4, with the focus being on 4D hypercomplex numbers. Section 5 concludes the paper and suggests directions for future research.

2. Numerical Systems Based on Real Numbers

The first nine numerical systems based on real numbers are known to exist and are presented in Figure 1. These are part of the Cayley–Dickson algebra, based on real numbers, and have different dimensions based on 2^n, where n denotes the number of the imaginary units in the corresponding numerical system.

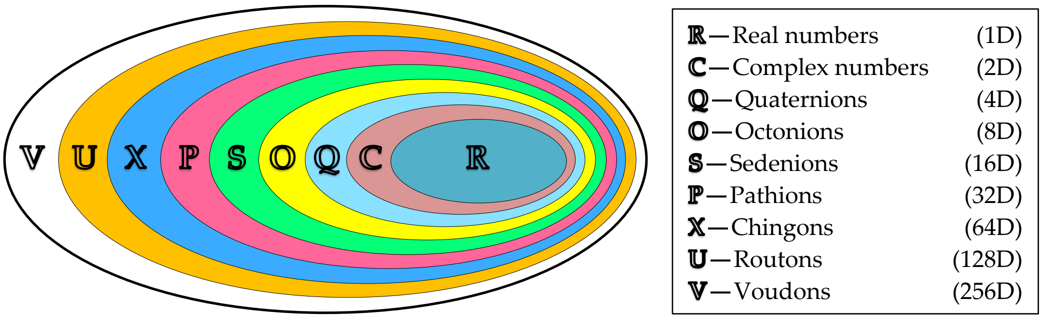

Figure 1. Numerical systems based on the real numbers with different dimensions, according to Cayley–Dickson algebra.

For $n = 0$, we obtain the *real numbers* (ℝ—Real), which are one-dimensional (1D) since $2^0 = 1$. For $n = 1$, $2^1 = 2$, the two-dimensional (2D) *complex numbers* (ℂ—Complex) are derived. For consistent terminology, 1D numbers have also been called *unarions*, and 2D numbers, *binarions* [1]. For $n = 2$, $2^2 = 4$—the numbers of dimension four (4D), *quaternions* (ℚ—Quaternions), are obtained, and for $n = 3$, $2^3 = 8$—the eight-dimensional (8D) *octonions* (𝕆—Octonions) are obtained. These are followed by the 16D *sedenions* (𝕊—Sedenions), 32D *pathions* (ℙ—Pathions), 64D *chingons* (𝕏—Chingons), 128D *routons* (𝕌—Routons), and 256D *voudons* (𝕍—Voudons). All number systems with a dimension higher than two are collectively called *hypercomplex numbers*.

The numbers of the first four numerical systems of the Cayley–Dickson algebra (Figure 1) are presented in Figure 2, where a and its associated subscript denote the real numbers, and i and the corresponding subscript denote the imaginary units.

a_0 - *real number*

$\downarrow$

$a_0 + a_1 i_1 = (a_0, a_1)$ - *complex number*

$\downarrow$

$a_0 + a_1 i_1 + a_2 i_2 + a_3 i_3 = (a_0, a_1, a_2, a_3)$ - *quaternion*

$\downarrow$

$a_0 + a_1 i_1 + a_2 i_2 + a_3 i_3 + a_4 i_4 + a_5 i_5 + a_6 i_6 + a_7 i_7 = (a_0, a_1, a_2, a_3, a_4, a_5, a_6, a_7)$ - *octonion*

Figure 2. Cayley–Dickson numerical systems represented by real numbers.

In this paper, real numbers will be denoted by the lower-case letters of the Latin alphabet, except for i, j, and k, which, with or without numerical subscripts, will be used to denote the imaginary units, forming the so-called base of the numerical system $\{1, i_1, i_2, \ldots, i_{\mathrm{n}}\}$. Usually, the base of the numerical system satisfies the condition: $i_l^2 \in \{-1,\ 0,\ +1\}$, i.e., the square of the imaginary unit can be equal to -1, to 0, or to +1.

As can be seen from Figure 2, the numbers of the different numerical systems can also be represented as ordered sequences of real numbers $(a_0, a_1, a_2, \ldots, a_n)$. To denote the numbers of the numerical systems with a higher dimension than the one-dimensional real numbers, capital Latin letters with or without a numerical subscript will be used, namely, for complex numbers—C, for quaternions—Q, for biquaternions—BQ, and for bicomplex numbers—BC.

2.1. Real (1D) and Complex (2D) Numbers

The one-dimensional system of real numbers $\mathbb{R}$ has been known to mankind since ancient times, although the name "real" was only officially introduced in the 17th century by the French mathematician René Descartes. One of the most significant hurdles associated with real numbers, which has taken centuries to overcome, is the adoption of negative real numbers. Besides their ability to realize counting and ordering, real numbers are also the means of expressing linear distance. Being one-dimensional, the real numbers constitute an infinite line, called the axis of real numbers, which the number zero divides into two semi-axes—for real positive numbers and for real negative numbers.

The complex numbers $\mathbb{C}$ owe their existence to the problem arising in the solution of the quadratic equation $x^2 + 1 = 0$. The problem with the cubic equation is similar in the case where all three of its roots are real numbers—they may contain the square root of a negative number that cannot be represented as a real number, which is why it is called an imaginary number [2]. Thus, in 1545, the Italian mathematician and physician Girolamo Cardano [3] arrived at the idea of complex numbers. Subsequently, for more than three centuries, many mathematicians worked on the theory of complex numbers—in 1794, the Swiss mathematician, physicist, and astronomer Leonard Euler defined the "imaginary unit", and in 1831, the German mathematician and physicist Johann Carl Friedrich Gauss definitively introduced the term "complex number" [4].

In modern mathematics, three types of numbers are defined by the 2D numerical system, depending on the value of the square of the imaginary unit i (Figure 3):

- ✓ *Complex numbers* for which $i^2 = -1$;
- ✓ *Hyperbolic numbers*, often called *split-complex numbers*, *double numbers*, or *bireal numbers*, for which $i^2 = +1$, but $i \neq \pm 1$;
- ✓ *Dual complex numbers*, when $i^2 = 0$, but $i \neq 0$.

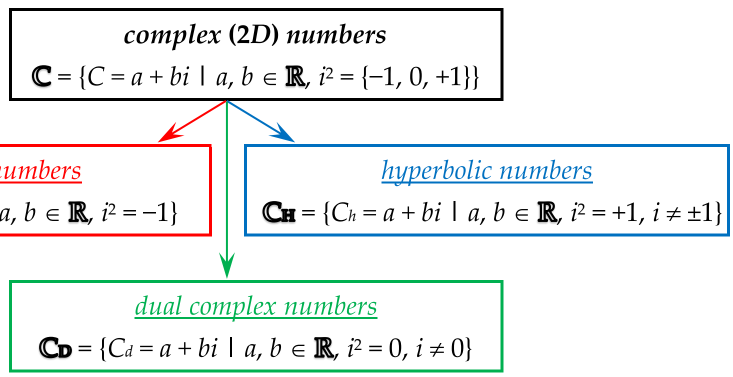

Figure 3. Types of complex (2D) numbers.

Each of these three types of complex number [5–7] is used as the basis for obtaining the corresponding type of higher dimension number.

Complex numbers implement all the operations typical of real numbers, as well as some impossible ones, such as modulus and complex conjugates. The dimensionality of complex numbers defines their domain of realization as points on an infinite plane called the complex plane (Figure 4).

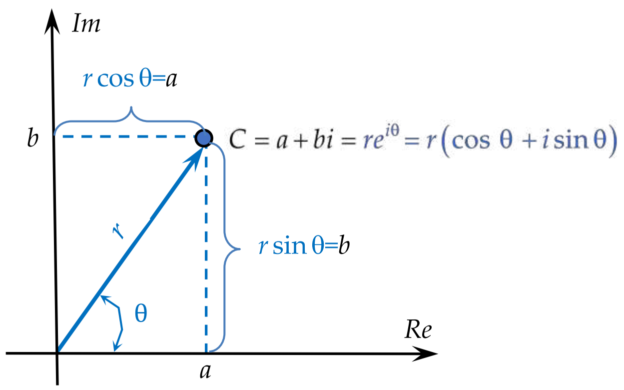

Figure 4. Geometric representation of the complex number $C = a + bi$ in Cartesian (rectangular) and polar coordinate systems.

The complex number C (Figure 4) can be represented in *arithmetic form* by its real and imaginary parts, as an ordered pair of real numbers, in *exponential form*, or in *trigonometric form* after the Euler transformation is used, [8,9]:

$$\begin{aligned} C &= a + bi \\ &= (a, b) \\ &= r\, e^{i\theta} \\ &= r(\cos\theta + i\sin\theta). \end{aligned} \tag{1}$$

$r = |C|$ is called the modulus (absolute value), and θ is the argument of the complex number, which can be determined by the real a and imaginary b parts of the complex number, as follows:

$$\begin{aligned} &r = \sqrt{a^2 + b^2} = |C| = \|C\|; \\ &\theta = arctg\tfrac{b}{a}. \end{aligned} \tag{2}$$

The modulus r of the complex number C is also called the Euclidean norm or canonical norm and is the length of the vector from the coordinate origin to the complex number (Figure 4).

The complex conjugate number of C is denoted by an asterisk and is:

$$\begin{aligned} C^* &= a - bi \\ &= r\, e^{-i\theta} \\ &= r(\cos\theta - i\sin\theta). \end{aligned} \tag{3}$$

The following dependencies hold for the complex numbers C_1, C_2, and C and its complex conjugate C^*:

$$\begin{aligned} &C + C^* = 2a; \quad C - C^* = -i2b; \\ &C^{-1} = \tfrac{C^*}{N(C)}; \quad \tfrac{C_1}{C_2} = \tfrac{C_1 C_2^*}{N(C_2)}. \end{aligned} \tag{4}$$

$N(C)$ is called the norm of the complex number and is the product of the complex number C and its complex conjugate number C^*:

$$N(C) = r^2 = a^2 + b^2 = |C|^2 = |C^*|^2 = C \cdot C^*. \tag{5}$$

The arithmetic operations, including multiplication of the complex number C by a numerical constant g, rooting, and exponentiation of a complex number are as follows:

$$C = g(a + bi) = (ga) + (gb)i. \tag{6}$$

$$\sqrt[n]{C} = \sqrt[n]{a + bi} = \sqrt[n]{r} \cdot e^{i\frac{\theta + 2k\pi}{n}}. \tag{7}$$

$$\begin{aligned} C^n = \quad & (a+bi)^n = r^n e^{in\theta} = r^n(\cos n\theta + i\sin n\theta); \\ & (a+bi)^2 = \left(a^2 - b^2\right) + i(2ab); \\ & (a+bi)^3 = \left(a^3 - 3ab^2\right) + i\left(3a^2b - b^3\right); \\ & (a+bi)^4 = \left(a^4 - 6a^2b^2 + b^4\right) + i\left(4a^3b - 4ab^3\right). \end{aligned} \tag{8}$$

The complex numbers $C_1 = a_1 + ib_1$ and $C_2 = a_2 + ib_2$ are equal ($C_1 = C_2$) if $a_1 = a_2$ and $b_1 = b_2$, which is equivalent to $r_1 = r_2$ and $\theta_1 = 2\pi k\theta_2$, $k = 0, \pm1, \pm2, \ldots$

The addition/subtraction, multiplication, and division operations of the complex numbers C_1 and C_2 are:

$$\begin{aligned} C_1 \pm C_2 \quad & = (a_1 + b_1 i) \pm (a_2 + b_2 i) \\ & = (a_1 \pm a_2) + (b_1 \pm b_2)i \\ & = r_1 e^{i\theta_1} \pm r_2 e^{i\theta_2} = (r_1\cos\theta_1 \pm r_2\cos\theta_2) + i(r_1\sin\theta_1 \pm r_2\sin\theta_2); \end{aligned} \tag{9}$$

$$\begin{aligned} C_1C_2 \quad & = (a_1 + b_1 i)(a_2 + b_2 i) \\ & = (a_1a_2 - b_1b_2) + (b_1a_2 + a_1b_2)i \\ & = r_1r_2e^{i(\theta_1+\theta_2)} = r_1r_2[\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)]; \end{aligned} \tag{10}$$

$$\begin{aligned} \tfrac{C_1}{C_2} \quad & = \tfrac{a_1+b_1 i}{a_2+b_2 i} \\ & = \tfrac{a_1a_2+b_1b_2}{a_2^2+b_2^2} + \tfrac{-a_1b_2+b_1a_2}{a_2^2+b_2^2} i \\ & = \tfrac{r_1}{r_2} e^{i(\theta_1-\theta_2)} = \tfrac{r_1}{r_2}[\cos(\theta_1-\theta_2) + i\sin(\theta_1-\theta_2)]. \end{aligned} \tag{11}$$

The theoretical basis of complex numbers is now fully developed in detail, and there have long been no theoretical obstacles to their widespread application in a variety of fields, including engineering. Complex numbers have been used in radar systems [10,11], in digital complex adaptive systems and digital filtering [12–14], in image processing [15,16], and in many other applications.

2.2. Quaternions (4D) and Octonions (8D)

In mathematics, the *quaternions* $\mathbb{Q}$ are an extension of complex number theory and represent a higher dimensional generalization of the two-dimensional complex numbers $\mathbb{C}$ (Figure 2). Only a few years after the full development of the algebra of complex numbers, in 1843, the Irish mathematician Sir William R. Hamilton defined quaternions as a generalization of complex numbers. In fact, some of the earliest work concerning quaternions and their properties belongs to Euler, but Hamilton is considered to be their true originator, having related the three imaginary units to the properties of rotation [17]. The *Lectures on Quaternions* published in 1853 [18] and *Elements of Quaternions* in 1866 [19] are among Hamilton's many notable works on the subject. It is also interesting to note that the German mathematician Gauss had likewise worked on the subject of quaternions a quarter of a century earlier. As early as 1819, Gauss worked on the subject of quaternions, but his work only gained recognition much later, in 1900. Many scientists, including Peter Tait [20], Josiah Willard Gibbs, Edwin Bidwell Wilson [21], Oliver Heaviside [22], and Hermann von Helmholtz, worked extensively on the theory and applications of Hamilton's quaternions and even founded the International Quaternion Association.

Broadly speaking, a *quaternion* is a group of four real numbers a, b, c, and d—one scalar a, and one tri-dimensional vector (b, c, d). Being a member of the large family of hypercomplex numbers, quaternions are often referred to, not entirely accurately, by the more general name *hypercomplex numbers*. The quaternion is represented in hypercomplex form as follows:

$$\begin{aligned} Q \quad & = a + bi_1 + ci_2 + di_3 \\ & = Q_S + Q_V, \end{aligned} \tag{12}$$

where $Q_S = a$ and $Q_V = bi_1 + ci_2 + di_3$ are called, respectively, the scalar and vector parts of the quaternion. a, b, c, and d, as noted, are real numbers, while i_1, i_2, and i_3 are mutually orthogonal imaginary units whose squares, in general, can be equal to -1, +1, or 0. Depending on this, and on the multiplication rules valid for i_1, i_2, and i_3, different types of quaternions are defined, all belonging to the family of 4D hypercomplex numbers. In Table 1, three of them, defined by Cantor and Solodovnikov [23], are given, which by no means exhausts all the possible combinations of values and mutual dependencies of imaginary units.

Table 1. Types of quaternion (4D) numbers.

$Q = a + bi_1 + ci_2 + di_3$		
a,b,c,d—Real Numbers		***i_1, i_2, i_3***—Imaginary Units
Types of Quaternions	***i_1, i_2, i_3***	**Canonical (Euclidean) Norm**
pseudo-quaternions	$i_1^2 = -1;\ {i_2}^2 = +1;\ i_3^2 = +1;$ $i_1 i_2 = i_3;\quad i_2 i_1 = -i_3;$ $i_2 i_3 = -i_1;\quad i_3 i_2 = i_1;$ $i_1 i_3 = -i_2;\quad i_3 i_1 = i_2.$	$\lVert Q \rVert = \sqrt{a^2 - b^2 - c^2 - d^2}$
degenerate quaternions	$i_1^2 = -1;\ {i_2}^2 = 0;\ i_3^2 = 0;$ $i_1 i_2 = i_3;\quad i_2 i_1 = -i_3;$ $i_2 i_3 = 0;\quad i_3 i_2 = 0;$ $i_1 i_3 = -i_2;\quad i_3 i_1 = i_2.$	$\lVert Q \rVert = \sqrt{a^2 + b^2}$
degenerate pseudo-quaternions	$i_1^2 = 1;\ {i_2}^2 = 1;\ i_3^2 = 0;$ $i_1 i_2 = i_3;\quad i_2 i_1 = -i_3;$ $i_2 i_3 = 0;\quad i_3 i_2 = 0;$ $i_1 i_3 = i_2;\quad i_3 i_1 = -i_2.$	$\lVert Q \rVert = \sqrt{a^2 - b^2}$

The most commonly used are Hamilton's quaternions, which we will consider in this publication and which we will call simply *quaternions*. For their imaginary units, known as the orthogonal basis, the following relations are satisfied:

$$\begin{gathered} i_1^2 = {i_2}^2 = i_3^2 = i_1 i_2 i_3 = -1; \\ i_1 i_2 = i_3;\ i_2 i_1 = -i_3; \\ i_2 i_3 = i_1;\ i_3 i_2 = -i_1; \\ i_3 i_1 = i_2;\ i_1 i_3 = -i_2. \end{gathered} \tag{13}$$

The vector part Q_V is also known as the imaginary part, and the scalar part Q_S as the real part of the quaternion. When the vector part is zero, the quaternion $Q = Q_S$ is called the scalar (real) quaternion, and when the scalar part is zero, $Q = Q_V$ is called the vector quaternion. When both parts are equal to zero, a zero quaternion results.

The conjugate quaternion of (12) is:

$$\begin{aligned} Q^* &= a - bi_1 - ci_2 - di_3 \\ &= Q_S - Q_V. \end{aligned} \tag{14}$$

The scalar Q_S and vector Q_V parts of the quaternion (12) can be obtained by the conjugated quaternion (14) as follows:

$$\begin{aligned} \tfrac{Q+Q^*}{2} &= Q_S = a; \\ \tfrac{Q-Q^*}{2} &= Q_V = bi_1 + ci_2 + di_3. \end{aligned} \tag{15}$$

The canonical (Euclidean) norm of a quaternion Q is the square root of its product with the conjugate quaternion Q^* and is a non-negative real number:

$$\|Q\| = \sqrt{QQ^*} = \sqrt{Q^*Q} = \sqrt{a^2 + b^2 + c^2 + d^2}. \tag{16}$$

The quaternion Q and its conjugate quaternion Q^* have the same real parts and the same canonical norms.

For the quaternions Q, Q_1, and Q_2, the reciprocal quaternion Q^{-1}, and the real constant g, the following equivalences are satisfied:

$$\|gQ\| = |g|\|Q\|; \quad \|Q_1Q_2\| = \|Q_1\|\|Q_2\|; \quad Q^{-1} = \frac{Q^*}{\|Q\|^2} = \frac{1}{Q}. \tag{17}$$

Similar to complex numbers, the norm of a quaternion Q is the square of its canonical norm (16):

$$N(Q) = \|Q\|^2 = a^2 + b^2 + c^2 + d^2. \tag{18}$$

Any quaternion can be represented by a so-called polar decomposition, using the so-called unit quaternion or versor U_Q, which is defined as a quaternion whose canonical norm is equal to one, i.e., $\|U_Q\| = 1$:

$$Q = U_Q\|Q\|. \tag{19}$$

An interesting property of a vector quaternion Q_V is that its square is equal to -1 if its norm is $\|Q_V\| = 1$. The set of all vector quaternions for which this property holds form a unit sphere and are called unit-length vector quaternions or right versors. Two vector quaternions are similar if $Q_{V1} = gQ_{V2}$, where g is a real constant.

There are simple rules for basic operations such as addition, subtraction, multiplication, and division by quaternions. Usually, only addition is meant, because subtraction is reduced to addition with a quaternion of the opposite sign. Addition is an associative and commutative operation and is performed by simple element-wise summation of the four real number components of the quaternions:

$$Q_1 + Q_2 = (a_1 + a_2) + (b_1 + b_2)i_1 + (c_1 + c_2)i_2 + (d_1 + d_2)i_3. \tag{20}$$

where:

$$\begin{aligned} Q_1 &= a_1 + b_1 i_1 \partial + c_1 i_2 + d_1 i_3; \\ Q_2 &= a_2 + b_2 i_1 + c_2 i_2 + d_2 i_3. \end{aligned} \tag{21}$$

The multiplication of the quaternion (12) by a numerical constant g is also an element-wise operation:

$$gQ = ga + gbi_1 + gci_2 + gdi_3. \tag{22}$$

The multiplication of quaternions is more complicated and is conducted by combinations of the components, subject to rules for the different vector products. The product of the quaternions $Q_1 = Q_{1S} + Q_{1V}$ and $Q_2 = Q_{2S} + Q_{2V}$, for example, is:

$$\begin{aligned} Q_1Q_2 &= (Q_1Q_2)_S + (Q_1Q_2)_V \\ &= (Q_{1S}Q_{2S} - Q_{1V} \bullet Q_{2V}) + (Q_{1S}Q_{2V} + Q_{2S}Q_{1V} + Q_{1V} \times Q_{2V}). \end{aligned} \tag{23}$$

The quaternion multiplication operation has the properties of associativity: $Q_1Q_2Q_3 = (Q_1Q_2)Q_3 = Q_1(Q_2Q_3)$, and distributivity: $Q_1(Q_2 + Q_3) = Q_1Q_2 + Q_1Q_3$, but not the property of commutativity: $Q_1Q_2 \neq Q_2Q_1$. The lack of commutativity in the multiplication of quaternions is due to the existence of a scalar product of the vector quaternions Q_{1V} and Q_{2V}:

$$Q_{1V} \bullet Q_{2V} = \frac{Q_{1V}^{*} Q_{2V} + Q_{2V}^{*} Q_{1V}}{2} = \frac{Q_{1V} Q_{2V}^{*} + Q_{2V} Q_{1V}^{*}}{2}, \tag{24}$$

and their vector product:

$$Q_{1V} \times Q_{2V} = \frac{Q_{1V} Q_{2V} - Q_{2V} Q_{1V}}{2}. \tag{25}$$

Like complex numbers, quaternions can implement the division operation using the norm (16) of the quaternion dividend Q_1 and the conjugate (14) of the quaternion divisor Q_2:

$$\frac{Q_1}{Q_2} = \frac{Q_1 Q_2^{*}}{\|Q_1\|}. \tag{26}$$

Hamilton's quaternion theory is a theory of vectors in space. The multiplication of quaternions corresponds to successive rotations of their vector parts about different axes and is thus closely related to the nature of three-dimensional space. A quaternion transforms one vector into another vector by multiplication. Representing spatial rotations by quaternions is much more compact and allows for faster computations compared to representing them by matrices. Quaternions are used in purely theoretical mathematics as well as in applied mathematics and physics, especially for efficient spatial rotation calculations [24–26]. They are of great importance for a number of modern applications such as biology, astronomy, mechanics, medicine, chemistry, geology, biophysics, etc.

When doubling the quaternion dimension, the 8D number system of octonions $\mathbb{O}$ is derived. One of the discoverers of octonions was another Irish mathematician and close associate of Hamilton—John Thomas Graves. His keen interest in Hamilton's work bore fruit and only a few months after the discovery of quaternions in 1843, Graves set out his idea for the eight-dimensional algebra of octonions, in correspondence with Hamilton, who congratulated him and promised to publish his discoveries. Unfortunately, this did not happen quickly enough and in the meantime, Arthur Cayley, a British mathematician also inspired by Hamilton's work, arrived at ideas similar to Graves' conclusions and published them [27]. Despite the objections of Graves, who did publish his work [28], but after Cayley, the accepted view is that it was Cayley who discovered eight-dimensional algebra, and octonions were hence given the name Cayley numbers.

Octonions are representable by real numbers (Figure 2) and have a double dimension compared to quaternions, i.e., they appear as their logical extension. If a_0, a_1, a_2, a_3, a_4, a_5, a_6, and a_7 are real coefficients, and $\{i_1, i_2, \ldots, i_7\}$ are imaginary units whose squares are equal to -1, the octonion can be represented as follows:

$$O = a_0 + a_1 i_1 + a_2 i_2 + a_3 i_3 + a_4 i_4 + a_5 i_5 + a_6 i_6 + a_7 i_7. \tag{27}$$

The dependencies of any three i_t, i_{t+1}, and i_{t+3} of the seven imaginary units satisfy the conditions (13) valid for the three imaginary units of the quaternion.

The operations addition, multiplication by a constant, and canonical norm of the octonion are equivalent to those of the quaternions. The division of octonions is also based on the norm and the conjugate octonion. Unlike numerical systems with lower dimensions, the multiplication of octonions is not associative.

According to the doubling dimension logic on which the Cayley–Dickson algebra is based, after octonions, the so-called sedenions $\mathbb{S}$ (16D number system) can be obtained, and so on (Figure 1). Most of the numerical systems presented in Figure 1 are still the focus of attention only in purely theoretical sciences. At present, the technical sciences are successfully working with real and complex numbers, and intensive research based on the

4D hypercomplex numbers, the quaternions, has been carried out in the last three or four decades. Octonions are not as well studied as complex numbers and quaternions, and as a result have not yet found widespread practical application. Octonions are associated with astonishing and unusual structures in mathematics, such as the extraordinary Lie groups, for example. The areas of application of octonions include string theory, quantum logic, and special relativity theory. The applications of octonions in physics are largely conjectural, such as attempts to study quarks via an octonic Hilbert space [29,30]. Octonions have been used to study the entropy and thermodynamics of black holes [31], and recent research areas include robotics, machine learning, neural networks, and image processing [32–35]. However, despite intensive research and publication activity, especially in the last few years, the full potential of the powerful mathematical apparatus of octonions is still to be unleashed.

Of particular importance for the practical application of numerical systems are the properties of the multiplication operation. The presence or absence of the properties of commutativity, associativity, alternativeness, and degree associativity for the first five number systems of the Cayley–Dickson algebra are given in Table 2.

Table 2. Properties of the multiplication operation in different numerical systems.

Number System	Dimension	Properties			
		Commutativity	Associativity	Alternativeness	Power Associativity
$\mathbb{R}$	1D	+	+	+	+
$\mathbb{C}$	2D	+	+	+	+
$\mathbb{Q}$	4D	−	+	+	+
$\mathbb{O}$	8D	−	−	+	+
$\mathbb{S}$	16D	−	−	−	+
commutativity—	$X_1X_2 = X_2X_1$				
associativity—	$X_1X_2X_3 = (X_1X_2)X_3 = X_1(X_2X_3)$				
alternativeness—	$(XX)Y = X(XY)$ or $Y(XX) = (YX)X$				
power associativity—	$XXX = (XX)X = X(XX)$				

X, X_1, X_2, X_3, and Y are elements of the corresponding number system—real numbers for $\mathbb{R}$, complex numbers for $\mathbb{C}$, quaternions for $\mathbb{Q}$, etc. The symbols "+" and "−" indicate the presence and absence of the corresponding property, respectively.

From Table 2, it can be clearly seen that each successive numerical system with higher dimensionality loses one of the properties of commutativity, associativity, alternativeness, and power associativity. Unlike the real numbers $\mathbb{R}$, the complex numbers $\mathbb{C}$ cannot express ordering, but both systems have all the listed properties. Quaternions $\mathbb{Q}$, like all consequent number systems, lose the commutativity property. The octonions $\mathbb{O}$, in addition to being noncommutative, are also nonassociative, but satisfy the alternativity property, which, as seen in Table 2, is a soft form of associativity. The sedenions $\mathbf{S}$, on the other hand, only manage to retain the property of power associativity.

3. Numerical Systems Based on Complex Numbers

Only the first four number systems $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Q}$, and $\mathbb{O}$ are *normed division algebras* based on the real numbers, i.e., spaces with multiplication and division. The multiplication operation has to be defined in such a way that it also allows the inverse operation, division. The norm, addition, and multiplication by a number for these four number systems are defined in an equivalent way. Multiplication by a nonzero element is a reversible operation, and for the norms of any two elements X and Y of any of these number systems, it is satisfied:

$$\|XY\| = \|X\|\|Y\|. \tag{28}$$

In addition to octonions, John Graves attempted to create a 16-dimensional normed division algebra, but without success. Each of the four normed division algebras is contained in the next, and each time losing one of the properties given in Table 2. With octonions, the associativity of multiplication is also lost, making it impossible, by doubling them, to create a new composition algebra. Sedenions exist as 16-dimensional structures, but the division operation cannot be defined for them. In 1898, the German mathematician Adolf Hurwitz proved that the only composition algebras are the four normed division algebras of the real numbers, complex numbers, quaternions, and octonions, i.e., the possible dimensions of a composition algebra are 1, 2, 4, and 8. Every composition algebra is an alternative algebra, i.e., the multiplication does not need to be associative, as it is enough to be alternative (Table 2) [36].

As already clarified, every normed division algebra has double the dimension of the previous one. The quaternions $\mathbb{Q}$, for example, can also be expressed in terms of the complex numbers $\mathbb{C}$: $\mathbb{Q} = \mathbb{C} + \mathbb{C}\, i$, and their dimension is twice that of the complex numbers. Therefore, in addition to being an ordered quaternion of real numbers (12), a quaternion can also be represented as an ordered pair of complex numbers:

$$\begin{aligned} Q &= a + bi_1 + ci_2 + di_3 \\ &= (a + bi_1) + i_2(c + di_1). \end{aligned} \tag{29}$$

If the first complex number is taken as the scalar part, then the quaternion represented in the form (29) can be called a *vector of complex numbers*. Similarly, octonions can be represented as an ordered pair of quaternions. This is the essence of the Cayley–Dickson principle of doubling the dimension of any number system with respect to the previous one, and is the only way to form a new normed algebra with division, i.e., by doubling the previous one.

The elements of each successive Cayley–Dickson numerical system of higher dimension (Figure 1) can also be obtained by replacing each real number in the preceding numerical system with a complex number—see Figure 5. This yields numerical systems based on complex numbers—bicomplex numbers, biquaternions, bioctonions, bisedenions, etc. It is obvious why they are also called *bi-number systems*.

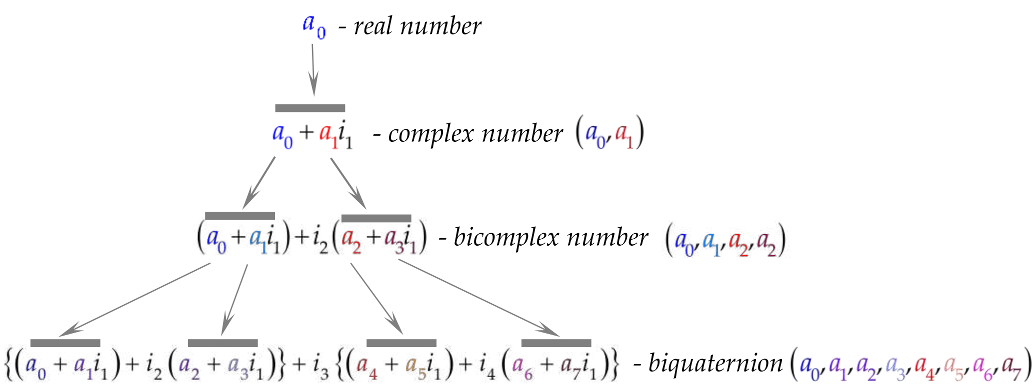

Figure 5. Numerical systems represented by complex numbers.

3.1. Bicomplex Numbers

The modern definition of bicomplex numbers describes them as an ordered pair of complex numbers C_1 and C_2 with specifically defined properties [37]:

$$BC = (a + bi_1) + i_2(c + di_1) = C_1 + i_2 C_2 = (C_1, C_2), \tag{30}$$

where $i_1{}^2 = \{-1, 0, +1\}$, $i_2{}^2 = \{-1, 0, +1\}$, and i_1, i_2, and $i_1 i_2$ are independent and commutative ($i_1 i_2 = i_2 i_1$). Therefore, bicomplex numbers are complex numbers with complex coefficients, which explains the name bicomplex.

The set $\{1, i_1, i_2, i_1 i_2\}$ is the basis of the bicomplex numbers, which can also be represented as four-dimensional numbers with real coefficients, as are the quaternions:

$$BC = a + bi_1 + ci_2 + di_1 i_2. \tag{31}$$

The need to find a solution to the problem of the missing commutativity of multiplication in quaternions (Table 2) led to the discovery of bicomplex numbers and the development of their theory. The first bicomplex numbers, called tessarines, were proposed almost at the same time as Hamilton's quaternions in 1848 by James Cockle, an English mathematician and lawyer [38]. This was prompted by irrational equations, such as $\sqrt{j} - 1 = 0$, whose roots are neither real nor complex. To obtain a solution to this problem, James Cockle proposed a new imaginary unit, whose square equals one, i.e., $j^2 = +1$ but $j \neq \pm 1$, and represents tessarines like quaternions:

$$T = a + bi_1 + ci_2 + di_3. \tag{32}$$

where a, b, c, and d are real numbers, and $i_1^2 = -1$, $i_2^2 = +1$, $i_1 i_2 = i_2 i_1 = i_3$. In a series of publications, he analyzed the algebraic properties of tessarines and compared them to other known hypercomplex numbers. Cockle generalized his discovery to higher dimensions, proposing octrines [39–41].

At the end of the 19th century, in 1892, the Italian mathematician Corrado Segre, studying complex geometry, proposed the concept of multicomplex numbers and the idea of an infinite set of n-complex algebras [42]. Among them is bicomplex algebra, which is obtained for $n = 2$. For $n = 4$, we speak of the biquaternion algebra (Figure 5), which had already been proposed by Hamilton some forty years earlier. Segre's bicomplex numbers can be represented either in the form (30) of pair of complex numbers or in the form (31) of four-dimensional numbers with real coefficients:

$$\begin{aligned} Segre's\ BC &= (a + bi_1) + i_2(c + di_1) \\ &= C_1 + i_2 C_2 \\ &= a + bi_1 + ci_2 + di_3, \end{aligned} \tag{33}$$

and the following dependencies are valid for their imaginary units:

$$i_1^2 = -1;\ i_2^2 = -1;\ i_3^2 = +1;\ i_3 = i_1 i_2 = i_2 i_1. \tag{34}$$

Bicomplex numbers have been studied for a long time, and to the name of Corrado Segre, those of Dragoni [43], Spampinato [44], and many other mathematicians and researchers can be added.

In an ordered pair (30), the complex numbers can be of any of the three types of complex numbers defined in Section 2.1 (Figure 3), namely, the classical complex numbers $\mathbb{C}_\mathbb{C}$, the hyperbolic numbers $\mathbb{C}_\mathbb{H}$, and the dual complex numbers $\mathbb{C}_\mathbb{D}$. Tessarines, for example, are hyperbolic numbers with complex coefficients. They, together with bicomplex numbers, are still of interest today [5,45–47]. It was only at the end of the 20th century that dual numbers with complex coefficients (complex dual numbers) were proposed [7], which, however, have noncommutative multiplication.

Depending on the specific values of the squares of the imaginary units, various bicomplex numbers are defined, some of the better known of which are presented in Table 3.

Table 3. Types of bicomplex (4D) numbers.

$\boldsymbol{BC} = (\boldsymbol{a} + \boldsymbol{b}\boldsymbol{i}_1) + \boldsymbol{i}_2(\boldsymbol{c} + \boldsymbol{d}\boldsymbol{i}_1) = \boldsymbol{C}_1 + \boldsymbol{i}_2\boldsymbol{C}_2 = \boldsymbol{a} + \boldsymbol{b}\boldsymbol{i}_1 + \boldsymbol{c}\boldsymbol{i}_2 + \boldsymbol{d}\boldsymbol{i}_3$ ***a, b, c, d***—Real Numbers; $\boldsymbol{C}_1$, $\boldsymbol{C}_2$—Complex Numbers; $\boldsymbol{i}_1$, $\boldsymbol{i}_2$, $\boldsymbol{i}_3$—Imaginary Units			
Type of Bicomplex Number	$\boldsymbol{i}_1$	$\boldsymbol{i}_2$	$\boldsymbol{i}_3 = \boldsymbol{i}_1\boldsymbol{i}_2 = \boldsymbol{i}_2\boldsymbol{i}_1$
bicomplex numbers	$i_1^2 = -1$	$i_2^2 = -1$	$i_3^2 = +1$
tessarines	$i_1^2 = -1$	$i_2^2 = +1;\ i_2 \neq \pm 1$	$i_3^2 = +1$
complex hyperbolic numbers	$i_1^2 = -1$	$i_2^2 = +1;\ i_2 \neq \pm 1$	$i_3^2 = -1$
hyperbolic complex numbers	$i_1^2 = +1;\ i_1 \neq \pm 1$	$i_2^2 = -1$	$i_3^2 = -1$
hyperbolic dual numbers	$i_1^2 = +1;\ i_1 \neq \pm 1$	$i_2^2 = 0;\ i_2 \neq 0$	$i_2 i_3 = 0;\ i_1 i_3 = i_2$
dual hyperbolic numbers	$i_1^2 = 0;\ i_1 \neq 0$	$i_2^2 = +1;\ i_2 \neq \pm 1$	$i_2 i_3 = i_1;\ i_1 i_3 = 0$
hyper-dual numbers	$i_1^2 = 0;\ i_1 \neq 0$	$i_2^2 = 0;\ i_2 \neq 0$	$i_3^2 = 0;\ i_3 \neq 0$

The bicomplex numbers (31) resemble Hamilton's quaternions (12), but in fact, these two types of 4D hypercomplex numbers differ substantially. It has already been mentioned that, unlike quaternions, the multiplication of bicomplex numbers is commutative. Achieving this commutativity has a cost, which is the appearance of so-called *zero divisors*, because of which the field of bicomplex numbers is not actually a field or a plane, but a ring with two imaginary units (i_1 and i_2) and can be seen as a combination of two complex spaces. A zero divisor is a nonzero element whose product with another nonzero element is zero. James Cockle called bicomplex zero divisors impossible, and their existence is the reason the field of bicomplex numbers is not integral. Bicomplex zero divisors are defined as follows: $\left\{ BC | BC \neq 0, C_1^2 + C_2^2 = 0 \right\}$. Among them, there are two (e_0 and $e_0{}^+$) of special importance, which are called *mutually complementary idempotent elements*, for which $e_0 = (e_0)^2$ and $e_0{}^+ = (e_0{}^+)^2$ are satisfied, as well as:

$$\left.\begin{array}{l} e_0 = \frac{1+i_1 i_2}{2} \\ e_0^+ = \frac{1-i_1 i_2}{2} \end{array}\right\} \begin{array}{l} e_0 = e_0^2 = e_0^{*+}; \quad e_0^+ = \left(e_0^+\right)^2 = \left(e_0^+\right)^{*+}; \\ e_0 + e_0^+ = 1; \quad e_0 \cdot e_0^+ = e_0^+ \cdot e_0 = 0. \end{array} \tag{35}$$

Any bicomplex number can also be expressed in terms of bicomplex zero divisors e_0 and $e_0{}^+$ (35), as follows [48]:

$$\begin{aligned} BC &= (a + bi_1) + i_2(c + di_1) \\ &= C_1 + i_2 C_2 = (C_1 - i_1 C_2)e_0 + (C_1 + i_1 C_2)e_0^+. \end{aligned} \tag{36}$$

By analogy with the complex conjugate number C^* (3), and due to the availability of three independent imaginary units, three different conjugates can be defined for a bicomplex number (33) [49]:

$$\begin{array}{l} BC^+ = (a + bi_1) - i_2(c + di_1) = C_1 - i_2 C_2; \\ BC^* = (a - bi_1) + i_2(c - di_1) = C_1^* + i_2 C_2^*; \\ BC^{*+} = (a - bi_1) - i_2(c - di_1) = C_1^* - i_2 C_2^*. \end{array} \tag{37}$$

All three bicomplex conjugates (37) satisfy the properties of additivity:

$$\begin{array}{c} (BC_1 + BC_2)^+ = BC_1^+ + BC_2^+; \\ (BC_1 + BC_2)^* = BC_1^* + BC_2^*; \\ (BC_1 + BC_2)^* + = BC_1^{*+} + BC_2^{*+}, \end{array} \tag{38}$$

involutivity:

$$\left(BC^*\right)^* = BC; \quad \left(BC^+\right)^+ = BC; \quad \left(BC^{*+}\right)^{*+} = BC, \tag{39}$$

and multiplicity:

$$\begin{aligned}(BC_1 \cdot BC_2)^+ &= BC_1^+ \cdot BC_2^+;\\ (BC_1 \cdot BC_2)^* &= BC_1^* \cdot BC_2^*;\\ (BC_1 \cdot BC_2)^{*+} &= BC_1^{*+} \cdot BC_2^{*+}.\end{aligned} \tag{40}$$

Regarding the multiplication of the bicomplex number BC with each of its three conjugates (37), the following relations are met:

$$BC \cdot BC^+ = C_1^2 + C_2^2, \tag{41}$$

$$BC \cdot BC^* = \left(|C_1|^2 - |C_2|^2\right) + 2\mathrm{Re}(C_1 C_2^*) i_2, \tag{42}$$

$$BC \cdot BC^{*+} = \left(|C_1|^2 + |C_2|^2\right) + 2\mathrm{Im}(C_1 C_2^*) i_1 i_2. \tag{43}$$

The norm of the bicomplex number (33) is:

$$N(BC) = BC^+ \cdot BC = (C_1 - i_2 C_2)(C_1 + i_2 C_2) = C_1^2 + C_2^2. \tag{44}$$

Since none of the modules in Formulas (42) and (43) are real numbers, the bicomplex number canonical (Euclidean) norm is defined as follows:

$$\|BC\| = \sqrt{C_1^2 + C_2^2} = \sqrt{a^2 + b^2 + c^2 + d^2}. \tag{45}$$

The canonical norm of the multiplication of two bicomplex numbers satisfies:

$$\|BC_1 \cdot BC_2\| \le \sqrt{2}\|BC_1\| \cdot \|BC_2\|. \tag{46}$$

Bicomplex numbers can be added and multiplied:

$$\begin{aligned}BC_1 + BC_2 &= (C_1 + i_2 C_2) + (C_3 + i_2 C_4)\\ &= (C_1 + C_3) + i_2(C_2 + C_4);\end{aligned} \tag{47}$$

$$\begin{aligned}BC_1 \cdot BC_2 &= (C_1 + i_2 C_2) \cdot (C_3 + i_2 C_4)\\ &= (C_1 C_3 - C_2 C_4) + i_2(C_1 C_4 + C_2 C_3).\end{aligned} \tag{48}$$

Both the sum (47) and the multiplication (48) of bicomplex numbers are commutative:

$$\begin{aligned}&BC_1 + BC_2 = BC_2 + BC_1;\\ &BC_1 \cdot BC_2 = BC_2 \cdot BC_1,\end{aligned} \tag{49}$$

associative:

$$\begin{aligned}&BC_1 + (BC_2 + BC_3) = (BC_1 + BC_2) + BC_3;\\ &BC_1 \cdot (BC_2 \cdot BC_3) = (BC_1 \cdot BC_2) \cdot BC_3,\end{aligned} \tag{50}$$

and the multiplication is also distributive:

$$BC_1 \cdot (BC_2 + BC_3) = BC_1 \cdot BC_2 + BC_1 \cdot BC_3 = (BC_2 + BC_3) \cdot BC_1. \tag{51}$$

Unlike complex numbers, in the domain of bicomplex numbers there are non-invertible elements, non-trivial elements, and non-trivial zero divisors.

Bicomplex numbers are the commutative version of quaternions, and when in a quaternion, the real coefficients are replaced by complex numbers and the so-called biquaternions are obtained.

3.2. Biquaternions

Along with quaternions, in 1853 Hamilton defined the so-called *biquaternions* [18]. These are derived from quaternions following the same logic by which bicomplex numbers are derived from complex ones. Biquaternions are 8D with respect to real numbers, but are 4D based on complex numbers (Figure 5), and are also called quaternions with complex coefficients. A biquaternion is an ordered group of four numbers C_0, C_1, C_2, and C_3, but each of them is complex:

$$\begin{aligned} BQ &= C_0 + i_1C_1 + i_2C_2 + i_3C_3 \\ &= C_0 + (i_1C_1 + i_2C_2 + i_3C_3) = BQ_S + BQ_V, \end{aligned} \tag{52}$$

where $C_l = a_l + ib_l$, $l = 0 \div 3$. By analogy with quaternions, for biquaternions, $BQ_S = C_0$ is called biscalar, and $BQ_V = i_1C_1 + i_2C_2 + i_3C_3$ is called bivector.

There is a commutativity between each of the imaginary units of the biquaternion (i_1, i_2, and i_3) and the imaginary unit i of its constituent complex numbers:

$$ii_1 = i_1i,\ ii_2 = i_2i,\ ii_3 = i_3i. \tag{53}$$

Three types of biquaternions are defined depending on the properties of the imaginary unit i. Ordinary biquaternions, also known as Hamiltonian biquaternions, are obtained when $i^2 = -1$, i.e., i is the classical imaginary unit. For split-biquaternions, also called Clifford biquaternions, $i^2 = 1$ (but $i \neq \pm 1$), and for dual quaternions $i^2 = 0$ (but $i \neq 0$).

As with quaternions, the multiplication of biquaternions is also not commutative, but can become zero, i.e., nonzero biquaternions that are zero divisors exist.

The biquaternion (52) $BQ = BQ_S + BQ_V$ has two conjugate forms: the bicomplex-conjugated biquaternion:

$$\begin{aligned} BQ^* &= C_0 - i_1C_1 - i_2C_2 - i_3C_3 \\ &= C_0 - (i_1C_1 + i_2C_2 + i_3C_3) = BQ_S - BQ_V, \end{aligned} \tag{54}$$

and the complex-conjugated biquaternion:

$$BQ^+ = C_0^+ + i_1C_1^+ + i_2C_2^+ + i_3C_3^+, \tag{55}$$

where $C_l = a_l + ib_l$ and $C_l{}^+ = a_l - ib_l$ for $l = 0 \div 3$.

For the two conjugate forms of the biquaternion in (54) and (55), the following dependencies hold:

$$\begin{aligned} &(BQ_1BQ_2)^* = BQ_2^*BQ_1^*; \\ &(BQ_1BQ_2)^+ = BQ_1^+BQ_2^+; \\ &(BQ^*)^+ = (BQ^+)^*. \end{aligned} \tag{56}$$

Biquaternions were introduced to the scientific community in the 19th century, but their full mathematical theory was not completed until a century later. Part of it was the idea that biquaternions could be obtained by bicomplex numbers, just as quaternions can be obtained by complex numbers, subject to the Cayley–Dickson principle of doubling the dimension of each number system regarding the previous one. According to this principle, the complex numbers are, for convenience, represented as ordered pairs of real numbers:

$$\begin{aligned} C_1 &= a_1 + ib_1 = (a_1, b_1), \\ C_2 &= a_2 + ib_2 = (a_2, b_2), \\ C_3 &= a_3 + ib_3 = (a_3, b_3), \\ C_4 &= a_4 + ib_4 = (a_4, b_4), \end{aligned} \tag{57}$$

and two by two can form a bicomplex number:

$$\begin{aligned} BC_1 &= (C_1, C_2) = (a_1, b_1, a_2, b_2); \\ BC_2 &= (C_3, C_4) = (a_3, b_3, a_4, b_4). \end{aligned} \tag{58}$$

Then, the biquaternion BQ, in turn, will be an ordered pair of the two bicomplex numbers (58), or an ordered quaternion of the complex numbers (57):

$$\begin{aligned} BQ &= (BC_1, BC_2) \\ &= (C_1, C_2, C_3, C_4). \end{aligned} \tag{59}$$

The multiplication of two biquaternions $BQ_1 = (BC_1, BC_2)$ and $BQ_2 = (BC_3, BC_4)$ will be:

$$\begin{aligned} BQ_1BQ_2 &= (BC_1, BC_2)(BC_3, BC_4) \\ &= \left(BC_1BC_3 - BC_4^*BC_2, BC_4BC_1 + BC_2BC_3^*\right). \end{aligned} \tag{60}$$

The conjugate of the biquaternion BQ, represented as an ordered pair of bicomplex numbers (59), can also be represented as follows, in addition to the form in (54):

$$BQ^* = (BC_1, BC_2)^* = (BC_1{}^*, -BC_2). \tag{61}$$

The norm of the biquaternion BQ (52) is:

$$N(BQ) = C_0^2 + C_1^2 + C_2^2 + C_3^2. \tag{62}$$

and for the norms of every two quaternions, BQ_1 and BQ_2 are fulfilled:

$$N(BQ_1BQ_2) = N(BQ_1)N(BQ_2). \tag{63}$$

Biquaternions also find an application in technical and engineering fields. Thus, just as single quaternions are used to represent rotation in three-dimensional space, biquaternions are a very convenient means of describing arbitrary 3D motion. The application areas of biquaternions include robotics, neural networks, computer vision, and computer graphics [50–52].

4. Four-Dimensional (4D) Numerical Systems Suitable for DSP

The fact that the multiplication operation of many hypercomplex numbers, such as the Hamiltonian quaternions and biquaternions, for example, does not have the commutativity property makes these hypercomplex numbers difficult to apply in the field of DSP. Currently, no digital processing algorithms based on noncommutative algebra have been developed. This makes only commutative, in terms of multiplication, numerical systems suitable for DSP applications. Only for these numerical systems is the convolution rule in the hypercomplex case in the time- and z-domains

$$Y(z) = H(z)X(z) \leftrightarrow y(n) = h(n) * x(n), \tag{64}$$

valid. In general, algebras with applications in DSP should have both associativity and commutativity properties (Table 4).

Table 4. Properties of complex and hypercomplex algebras.

	Hypercomplex Algebras					
Property	$\mathbb{C}$	$\mathbb{Q}$	$\mathbb{BQ}$	$\mathbb{RBQ}$	$\mathbb{BC}$	$\mathbb{T}$
Associativity $a(bc) = (ab)c$	+	+	+	+	+	+
commutativity $ab = ba$	+	−	−	+	+	+
convolution $h(n){*}x(n) \leftrightarrow H(z)X(z)$	+	−	−	+	+	+
zero divisors $ab = 0; b \neq 0; a \neq 0$	−	−	+	+	+	+
normed division algebra	+	+	−	−	−	−
norm $N(\ )$	+	+	+	+	+	+
canonical norm (Euclidean) $\sqrt{N(\ )} = \|\ \| = \lvert\ \rvert$	+	+	+	+	+	+

$\mathbb{C}$—complex; $\mathbb{Q}$—quaternions; $\mathbb{BQ}$—biquaternions; $\mathbb{RBQ}$—reduced biquaternions; $\mathbb{BC}$—bicomplex; and $\mathbb{T}$—tessarines.

Such algebras, of dimension 4D and higher, have zero divisors, which is valid for associative, but not commutative, algebras of dimension 8D and higher. From a DSP point of view, normed division algebras are neither required nor recommended because they lack zero divisors. The presence of zero divisors can be seen as an advantage since it allows orthogonal decomposition which, in turn, facilitates the analysis of digital systems and improves computational efficiency.

Scientific results have been reported on which hypercomplex number systems of what dimension would be suitable for DSP [23,46,53–56]. As has become evident from the exposition so far, the higher the dimension of a numerical system, the more types of hypercomplex algebras that can be defined in it.

Among the 4D numerical systems based on complex numbers, only those representing an ordered pair of classical complex numbers $\mathbb{C}_{\mathbb{C}}$ ($i^2 = -1$) or an ordered pair of hyperbolic numbers $\mathbb{C}_{\mathbb{H}}$ ($i^2 = +1$) are commutative—see Figure 6.

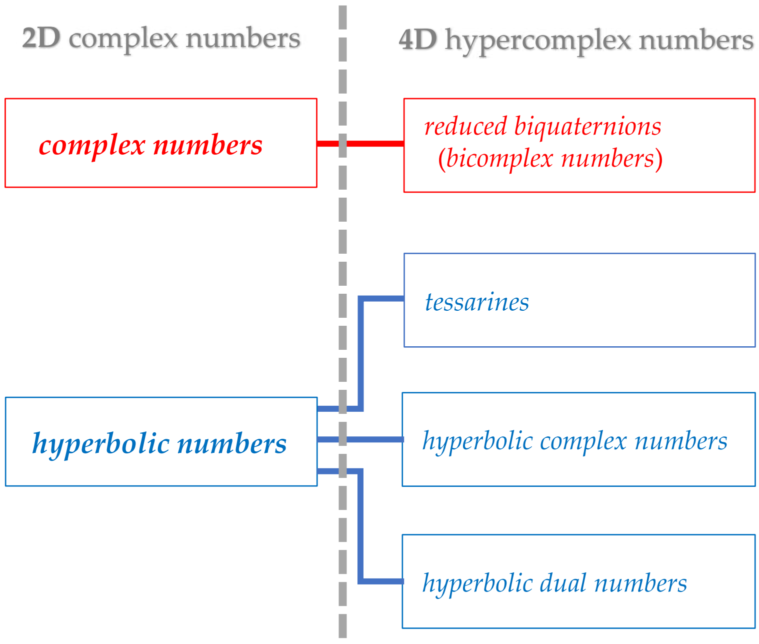

Figure 6. The 2D and 4D numbers with commutative algebra.

Modern digital algorithms basically use real numbers. Complex numbers are also well suited to represent signals and systems because they allow the compact implementation and appropriate interpretation of quite complex processing [13,14]. Bicomplex numbers contribute even more to the compact implementation of the signals and advanced DSP algorithms due to their mathematical properties. The presence of zero divisors and the possibility for them to be represented in exponential and trigonometric form contribute to the simplification of the signal processing, thus improving the DSP computational efficiency [46,54–56].

In this work, we will take note of the 4D hypercomplex numbers named reduced biquaternions ($\mathbb{RBQ}_\mathbf{S}$), which, in fact, are bicomplex numbers, and this name has been used in a number of research works [37,45,48,49,56–58]. It is evident from Table 4 that the properties of reduced biquaternions ($\mathbb{RBQ}$) and bicomplex numbers ($\mathbb{BC}$) are identical.

4.1. Reduced Biquaternions

The two main advantages of quaternions $\mathbb{Q}$ are that they have a clear and unambiguous geometric meaning and that they form a normed division algebra in which the division operation (except to zero) is always possible. The impossibility of using them in DSP, however, leads to the emergence of reduced biquaternions $\mathbb{RBQ}_\mathbf{S}$ as their alternative. Together with the fact that $\mathbb{RBQ}_\mathbf{S}$ implement commutative multiplication, they have another important property—for them, the hypercomplex convolution (64) is valid, which can also be represented in the frequency domain by the Discrete Fourier Transform (DFT).

Unlike quaternions, which represent rotation in three-dimensional space perfectly, reduced biquaternions have a complex geometric meaning that is useful for relativity but not for engineering [53].

It is accepted that the name *reduced biquaternions* was proposed by Hans-Dieter Schutte and Jorg Wenzel [59], who also introduced a new modified numerical system which is commutative with respect to multiplication. According to them, the reduced biquaternion can be obtained from the quaternion (12) if the following two actions are performed: (1) the coefficients c and d are set to zero, which will effectively reduce the quaternion to a complex number; and (2) the real coefficients a and b are expanded to complex numbers. This procedure is equivalent to replacing each of the two real coefficients of a complex number with complex numbers, resulting in a bicomplex number. Obviously, the reduced biquaternion is inherently a bicomplex number since it is an ordered pair of complex numbers and as such belongs to a 4D numerical system.

So, if we replace the real numbers a and b, representing the real and imaginary parts of the complex number C, respectively, by two complex numbers, we obtain the reduced biquaternion A [59]:

$$\begin{aligned} C = a + ib \Rightarrow \quad & (a_0 + ja_1) + i(a_2 + ja_3) \\ & = A_S + iA_V \\ & = a_0 + ja_1 + ia_2 + ka_3 = A, \end{aligned} \tag{65}$$

where a_0, a_1, a_2, and a_3 are real numbers and the complex numbers A_S and A_V are called, respectively, the scalar and vector parts of the reduced biquaternion. There are three imaginary units in the reduced biquaternion A, and in this paper, they will be denoted by j, i, and k.

j is the classical imaginary unit for which $j^2 = -1$ is satisfied. The i is called the vector unit such that $i^2 = -1$, while the square of the imaginary unit k is equal to +1. The properties that the three imaginary units of the reduced biquaternion A satisfy are as follows [59]:

$$\begin{aligned} i^2 = j^2 = -1; \quad k^2 = +1; \\ ij = ji = k; \quad jk = kj = -i; \quad ik = ki = -j \ . \end{aligned} \tag{66}$$

The comparison of (66) and (13) shows the presence of the commutativity of the imaginary units in reduced biquaternions, as opposed to quaternions, which, as already pointed out, is their main advantage in DSP applications.

Of importance for reduced biquaternions are their two idempotent zero divisors e_1 and e_2, which, by analogy with those introduced for bicomplex numbers (35), are:

$$e_1 = \frac{1+k}{2}; \quad e_2 = \frac{1-k}{2}. \tag{67}$$

e_1 and e_2 are both *idempotent* ($e_1 = {e_1}^2$ and $e_2 = {e_2}^2$) and zero divisors ($e_1 e_2 = 0$), and reduced biquaternions can be represented by them as well [53]:

$$\begin{aligned} A \ &= (a_0 + ja_1) + i(a_2 + ja_3) \\ &= A_S + iA_V \\ &= (A_S + A_V)e_1 + (A_S - A_V)e_2 \\ &= [(a_0 + a_2) + j(a_1 + a_3)]e_1 + [(a_0 - a_2) + j(a_1 - a_3)]e_2. \end{aligned} \tag{68}$$

The representation of the reduced biquaternion by the two zero divisors (67) leads to a reduction in computational complexity and the simplification of the analysis for operations such as multiplication and Fourier transform.

Reduced biquaternions can also be represented in exponential form:

$$\begin{aligned} A = \ & a_0 + ja_1 + ia_2 + ka_3 \\ & = \|A\| e^{j\theta_j} e^{i\theta_i} e^{k\theta_k}. \end{aligned} \tag{69}$$

$\|A\|$ is the canonical (Euclidean) norm of the reduced biquaternion A (65) and is defined as follows:

$$\|A\| = |A| = \sqrt[4]{\left(a_0^2 + a_1^2 + a_2^2 + a_3^2\right)^2 - 4(a_0 a_3 - a_2 a_4)^2} \geq 0. \tag{70}$$

The exponential representation (69) is possible if the canonical norm (70) is not zero and if the three polar phases θ_j, θ_i, and θ_k take values in the intervals:

$$\theta_j \in (-\pi, \pi]; \quad \theta_i \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right]; \quad \theta_k \in R. \tag{71}$$

As with complex numbers, exponential representation (69) is very suitable for DSP applications, especially if it is brought into trigonometric form using the Euler and Lorentz transformations, giving the relation of exponential functions, respectively, to the trigonometric and to the hyperbolic functions [6,9,53]. The trigonometric form of the reduced biquaternion is:

$$\begin{aligned} A \ &= \|A\| e^{j\theta_j} e^{i\theta_i} e^{k\theta_k} \\ &= \|A\| \left[\left(\cos\theta_j \cos\theta_i\right) + j\left(\sin\theta_j \cos\theta_i\right) + i\left(\cos\theta_j \sin\theta_i\right) + k\left(\sin\theta_j \sin\theta_i\right) \right]. \end{aligned} \tag{72}$$

The absence of θ_k in (72) is explained by the impossibility of applying Euler's formula to $e^{k\theta k}$, since k is not the classical imaginary unit ($k^2 = +1$) and, therefore, instead of the trigonometric functions sin and cos, the hyperbolic functions sinh and cosh and the Lorentz transformation are used:

$$e^{k\theta_k} = \cosh\theta_k + k\sinh\theta_k. \tag{73}$$

The link between the form that uses the zero divisors e_1 and e_2 (67) and the exponential form (69) can be derived through the following mathematical relationships:

$$\begin{aligned} e^{j\theta_j} &= e^{j\theta_j}e_1 + e^{j\theta_j}e_2 \\ e^{i\theta_i} &= e^{j\theta_i}e_1 + e^{-j\theta_i}e_2 \\ e^{k\theta_k} &= e^{\theta_k}e_1 + e^{-\theta_k}e_2 \end{aligned} \tag{74}$$

and is:

$$\begin{aligned} A &= \|A\|e^{j\theta_j}e^{i\theta_i}e^{k\theta_k} \\ &= \|A\|\Big[\Big(e^{\theta_k}e^{j(\theta_j+\theta_i)}\Big)e_1 + \Big(e^{-\theta_k}e^{j(\theta_j-\theta_i)}\Big)e_2\Big]. \end{aligned} \tag{75}$$

For the reduced biquaternion A, two types of conjugates are defined—the vector conjugate:

$$\begin{aligned} A^+ &= A_S - iA_V \\ &= (a_0 + ja_1) - i(a_2 + ja_3) \\ &= \|A\|e^{-j\theta_j}e^{-i\theta_i}e^{-k\theta_k}. \end{aligned} \tag{76}$$

and complex conjugate reduced biquaternion:

$$\begin{aligned} A^* &= A_S^* + iA_V^* \\ &= (a_0 - ja_1) + i(a_2 - ja_3). \end{aligned} \tag{77}$$

In addition, a so-called doubly conjugate reduced biquaternion is defined:

$$\begin{aligned} A^{*+} &= A_S^* - iA_V^* \\ &= (a_0 - ja_1) - i(a_2 - ja_3). \end{aligned} \tag{78}$$

The scalar A_S and vector A_V parts of A (65) can be determined by the vector conjugate reduced biquaternion A^+ (76) as follows:

$$A_S = \frac{A + A^+}{2}; \quad A_V = \frac{A - A^+}{2i}. \tag{79}$$

Normally, the norm is a real number, but the norm of the reduced biquaternion is defined in the domain of complex numbers and is as follows:

$$N(A) = \|A\|^2 = AA^+ = (A_S + iA_V)(A_S - iA_V). \tag{80}$$

When the norm of the reduced biquaternion is $N(A) \neq 0$, the so-called inverse or reciprocal reduced biquaternion can be defined:

$$A^{-1} = \frac{A^+}{N(A)} = \frac{e^{-j\theta_j}e^{-i\theta_i}e^{-k\theta_k}}{\|A\|}. \tag{81}$$

Analogously to (81), the reciprocal of the difference of two reduced biquaternions A and B, whose norm $N(A-B)$ is not zero, will be:

$$(A - B)^{-1} = \frac{A^+ - A^+}{N(A - B)}. \tag{82}$$

In contrast to complex numbers, for reduced biquaternions A and B and their vector conjugates A^+ and B^+, the following is satisfied:

$$(A+B)^+ \neq A^+ + B^+. \tag{83}$$

Let us define the basic arithmetic operations for the reduced biquaternions A and B:

$$\begin{aligned} A &= (a_0 + ja_1) + i(a_2 + ja_3) = \|A\| e^{j\theta_{jA}} e^{i\theta_{iA}} e^{k\theta_{kA}}; \\ B &= (b_0 + jb_1) + i(b_2 + jb_3) = \|B\| e^{j\theta_{jB}} e^{i\theta_{iB}} e^{k\theta_{kB}}. \end{aligned} \tag{84}$$

The addition and subtraction of the reduced biquaternions (84) are element-by-element operations:

$$A \pm B = [(a_0 \pm b_0) + j(a_1 \pm b_1)] + i[(a_2 \pm b_2) + j(a_3 \pm b_3)]. \tag{85}$$

The multiplication of the reduced biquaternions A and B (84) is:

$$\begin{aligned} AB &= \|A\|\|B\| e^{j(\theta_{jA}+\theta_{jB})} e^{i(\theta_{iA}+\theta_{iB})} e^{k(\theta_{kA}+\theta_{kB})} \\ &= D = (d_0 + jd_1) + i(d_2 + jd_3), \end{aligned} \tag{86}$$

where:

$$\begin{aligned} d_0 &= a_0b_0 - a_1b_1 - a_2b_2 + a_3b_3; \\ d_1 &= a_0b_1 - a_1b_0 - a_2b_3 + a_3b_2; \\ d_2 &= a_0b_2 - a_1b_3 - a_2b_0 + a_3b_1; \\ d_3 &= a_0b_3 - a_1b_2 - a_2b_1 + a_3b_0. \end{aligned} \tag{87}$$

The specifics of the represented actions and rules make reduced biquaternions one of the members of the family of bicomplex numbers and distinguish them from other members of the family by the basis {1, j, i, ji} used, the type of complex numbers in the ordered pair, and the values of the squares of their imaginary units. In 1992, T. A. Ell [60] defined so-called *double-complex algebra*, again with commutative multiplication, but unlike (66), the square of j is equal to +1. A set of 4D commutative hypercomplex algebras for which i = +1 is also proposed in [61].

4.2. Applications of 4D Numerical Systems in DSP

Over the last few decades, 4D hypercomplex numbers with commutative algebra have found increasing applications in various fields, such as power systems and energy efficiency [62–64]; biomedicine, bioengineering, and medical engineering [65–69]; meteorology [57,70,71]; energy and electronics [72–74]; aerospace engineering [51,75,76]; robotics [77–80], etc.

For the most part, technical sciences are associated with DSP. Signals and data are digitally processed to be improved or changed, and analyzed to determine specific informational content. DSP is applicable to both real-time data streams and static data sets and may involve linear or nonlinear operations. At present, the advantages of representing digital signals by complex and hypercomplex numbers and their accompanying digital processing are indubitable [46,54,55,58,59,81–88].

Digital signal processing is the essence of modern digital technologies in telecommunications. Modern 5G networks, distinguished by their heterogeneous nature and use of a wide frequency range, throw up many new scientific challenges that can be solved using the powerful mathematical apparatus of four-dimensional arithmetic. Among the telecommunication applications for which 4D hypercomplex processing is a successful and elegant tool for the description, analysis, and optimization of processes and systems, the following can be mentioned: audio and speech signal processing [15,16,53,89–91]; digital color image processing, restoration, and recognition [92–97]; noise and interference measurement and suppression [98–100]; video signal processing, pattern, emotion, and face recognition [101–105]; communication, neural, and adaptive networks [56,106–110]; data compression, coding, and information security [111–116]; radar, sonar, and sensor processing [10,117–121]; motion control systems, location, spatial orientation, and computer vision [50,122–126]; three-dimensional object reconstruction and localization [127–130], and many others. Due to their inevitable connection with digital processing of different types of signals, the listed applications involve hypercomplex processing, mostly based on Hamiltonian quaternions or reduced biquaternions.

5. Conclusions and Directions for Future Research

The present work outlines an overview of the numerical systems known so far and part of the Cayley–Dickson algebra, each one of which doubles the dimension of the previous one. Besides 1D real numbers and 2D complex numbers, those of higher dimensions are known as hypercomplex numbers.

The members of the large family of the hypercomplex numbers are both alike and yet very dissimilar. Due to their intrinsic and distinctive properties, hypercomplex numbers currently have numerous applications, most of which are referred to in this work. There is no doubt that the areas of application of hypercomplex numbers will continue to expand as this powerful mathematical tool increasingly attracts the attention of researchers.

As a general rule, the Cayley–Dickson numerical systems are based on real numbers, but another approach discussed in this paper suggests that hypercomplex numbers can be represented on the basis of complex numbers, thus introducing so-called bi-number systems—bicomplex, biquaternions, bioctonions, etc.

A significant part of the application of hypercomplex numbers relates to the technical sciences, which necessitate the formulation of hypercomplex numerical systems applicable to DSP. The 4D numerical systems suitable for DSP are outlined in this paper, with emphasis on bicomplex numbers which are ordered pairs of classical complex numbers, since these are the most widely used in today's advanced applications. They are often referred to as reduced biquaternions, and more detailed research and a comparison to the 4D numbers based on hyperbolic numbers (tessarines, hyperbolic dual numbers, etc.) will be beneficial for many DSP applications.

There are still many scientific challenges related to hypercomplex DSP that deserve to be more thoroughly and comprehensively analyzed and investigated, such as:

- Which of the possible representations of bicomplex numbers—as ordered pairs of complex numbers, as four-dimensional numbers with real coefficients, in terms of bicomplex zero divisors, in exponential form, or in trigonometric form—is the most appropriate for a particular application;
- Since the octonions are the last member of the normed division algebras, it is perhaps possible to develop commutative 8D numbers appropriate for DSP, just as quaternions were modified into commutative reduced biquaternions.

The problems listed above are only a part of possible future research related to hypercomplex numbers that can be used in digital signal processing algorithms.

As is often the case, solving a scientific challenge not only provides answers, but also poses many new questions at the same time. One thing is definite—DSP algorithms, real, complex, and bicomplex, have established themselves as an important part of the DSP scientific field and have found great practical application for more than 70 years. It is difficult to predict what technological innovations the near or more distant future will bring us, but to date it can be argued that the development and refinement of this type of digital processing is significant, promising, and noteworthy scientific pursuit.

Author Contributions: Conceptualization, writing—original draft preparation, visualization, supervision, Z.V.-J.; formal analysis, investigation, writing—review and editing, Z.V.-J. and D.M.; project administration, funding acquisition, M.N.; methodology, resources, validation, Z.V.-J., M.N., and D.M. All authors have read and agreed to the published version of the manuscript.

Funding: This study is financed by the European Union—NextGenerationEU, through the National Recovery and Resilience Plan of the Republic of Bulgaria, project No. BG-RRP-2.004-0005.

Data Availability Statement: Data is contained within the article; further inquiries can be directed to the corresponding author.

Acknowledgments: This work was supported by the research project BG-RRP-2.004-0005 "A multivalent Neural Network-based Cybersecurity System for Communications and Industrial Networks" of the operational program "National plan for recovery and sustainability", investment from the "Program for accelerating economic recovery and transformation through science and innovation", and the operation "Creation of a network of research universities in Bulgaria".

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. McCrimmon, K. *A Taste of Jordan Algebras*; Springer: Dordrecht, The Netherlands, 2004; ISBN 0-387-95447-3.
2. Descartes, R. *The Geometry of René Descartes*; with a Facsimile of the First Edition, Retrieved 20 April 2011; Dover Publications: Mineola, NY, USA, 1954; ISBN 978-0-486-60068-0.
3. Kline, M. *; Mathematical Thought from Ancient to Modern Times: Volume 1*; Oxford University Press: New York, NY, USA, 1972; ISBN 0195061357.
4. Gauss, C.F. *Theoria Residuorum Biquadraticorum*; Commentatio Secunda; Commentationes Societatis Regiae Scientiarum Gottingensis Recentiores: Gottingae, Germany, 1831.
5. Akar, M.; Yüce, S.; Şahin, S. On the Dual Hyperbolic Numbers and the Complex Hyperbolic Numbers. *J. Comput. Sci. Comput. Math.* **2018**, *8*, 1–6. [CrossRef]
6. Rooney, J. On the tree types of complex numbers and planar transformations. *Environ. Plan.* **1978**, *B5*, 89–99. [CrossRef]
7. Cheng, H.H.; Thompson, S. Dual Polynomials and Complex Dual Numbers for Analysis of Spatial Mechanisms. In Proceedings of the ASME 24th Biennial Mechanisms Conference, Irvine, CA, USA, 18–22 August 1996; pp. 19–22.
8. Euler, L. *Introductio in Analysin Infinitorum (Introduction to the Analysis of the Infinite)*; Marc Michel Bosquet & Co.: Lucerne, Switzerland, 1748; Volume 1, p. 104.
9. Yaglom, I.M. *Complex Numbers in Geometry*; Academic: New York, NY, USA, 1968.
10. Jouny, I.; Moses, R.L.; Garber, F.D. Classification of Radar Signals Using the Bispectrum. In Proceedings of the IEEE International Conference on Acoustics, , Speech, and Signal Processing (ICASSP'91), Toronto, ON, Canada, 14–17 May 1991; Volume 5, pp. 3429–3432.
11. Germann, U.; Boscacci, M.; Clementi, L.; Gabella, M.; Hering, A.; Sartori, M.; Sideris, I.V.; Calpini, B. Weather radar in complex orography. *Remote. Sens.* **2022**, *14*, 503. [CrossRef]
12. Pun, K.P.; Franca, J.E.; Azeredo-Leme, C. The Correction of Frequency-dependent I/Q Mistmatches in Quadrature Receivers by Adaptive Signal Separation. In Proceedings of the IEEE International Symposium on Circuits and Systems (ISCAS'01), Sydney, Australia, 6–9 May 2001; Volume 1, pp. I-424–I-427.
13. Nikolova, Z.; Stoyanov, G.; Iliev, G.; Poulkov, V. Complex Coefficient IIR Digital Filters. In *Digital Filters*; Márquez, F.P.G., Ed.; InTech: London, UK, 2011; pp. 209–239; ISBN 978-953-307-190-9. [CrossRef]
14. Nikolova, Z.; Iliev, G.; Ovtcharov, M.; Poulkov, V. Complex Digital Signal Processing in Telecommunications. In *Applications of Digital Signal Processing*; Cuadrado-Laborde, C., Ed.; InTech: London, UK, 2011; pp. 3–24; ISBN 978-953-307-406-1. [CrossRef]
15. Pun, K.P.; Franca, J.E.; Azeredo-Leme, C.; Reis, R. Ouadrature Sampling Schemes with Improved Image Rejection. *IEEE Trans. Circuits Syst.—II Analog. Digit. Signal Process.* **2003**, *50*, 641–647.
16. Gonzales, A.M.; Grigoryan, A.M. Fast Retinex for Color Image Enhancement: Methods and Algorithms. In Proceedings of the Conference on Mobile Devices and Multimedia: Enabling Technologies, Algorithms, and Applications, San Francisco, CA, USA, 10–11 February 2015; ISBN 978-1-62841-501-8, ISSN 0277–786X.
17. Hamilton, W.R. On Quaternions; or on a new System of Imaginaries in Algebra, Letter to John T. Graves. 17 October 1843. *Philos. Mag.* **1843**, *25*, 489–495.
18. Hamilton, W.R. *Lectures on Quaternions*; Royal Irish Academy: Dublin, Ireland, 1853.
19. Hamilton, W.R. *Elements of Quaternions*; Longmans, Green, & Co.: London, UK, 1866.

20. Tait, P.G. *An Elementary Treatise on Quaternions*, 2nd ed.; Cambridge University Press: Cambridge, UK, 1873.
21. Gibbs, J.W.; Wilson, E.B. *Vector Analysis*; Yale University Press: New Haven, CT, USA, 1901; p. 428.
22. Heaviside, O. *Electromagnetic Theory*; The Electrician Printing and Publishing Company: London, UK, 1893; pp. 134–135.
23. Кантор, И.Л.; Солодовников, А.С. *Гиперболические Числа*; изд. Наука: Москва, Russia, 1973.
24. Ward, J.P. *Quaternions and Cayley Numbers: Algebra and Applications*; Springer Science & Business Media: Berlin/Heidelberg, Germany, 2012.
25. McAulay, A. *Utility of Quaternions in Physics*; Holistence Publications: Çanakkale, Turkey, 2024.
26. Voight, J. *Quaternion Algebras*; Springers: Berlin/Heidelberg, Germany, 2021; ISSN 0072-5285.
27. Cayley, A. On Jacobi's elliptic functions and on quaternions. *J. Comput. Educ.* **1845**, *26*, 208–211. [CrossRef]
28. Graves, J.T. On a Connection between the General Theory of Normal Couples and the Theory of Complete Quadratic Functions of Two Variables. *J. Comput. Educ.* **1845**, *26*, 315–320. [CrossRef]
29. Günaydin, M.; Gürsey, F. Quark structure and octonions. *J. Math. Phys.* **1973**, *14*, 1651–1667. [CrossRef]
30. Günaydin, M.; Gürsey, F. Quark statistics and octonions. *Phys. Rev.* **1974**, *9*, 3387–3391. [CrossRef]
31. Borsten, L.; Dahanayake, D.; Duff, M.J.; Ebrahim, H.; Rubens, W. Black holes, qubits and octonions. *Phys. Rep.* **2009**, *471*, 113–219. [CrossRef]
32. Wu, J.; Xu, L.; Wu, F.; Kong, Y.; Senhadji, L.; Shu, H. Deep octonion networks. *Neurocomputing* **2020**, *397*, 179–191. [CrossRef]
33. Gao, H.-Y.; Lam, K.-M. From quaternion to octonion: Featurebased image saliency detection. In Proceedings of the ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing, Florence, Italy, 4–9 May 2014; pp. 2808–2812.
34. Popa, C.A. Stability and synchronization of octonion-valued neural networks with leakage and mixed delays on time scales. *Comput. Appl. Math.* **2024**, *43*, 300. [CrossRef]
35. Tahiri, M.A.; Boudaaoua, B.; Karmouni, H.; Tahiri, H.; Oufettoul, H.; Amakdouf, H.; Qjidaa, H.; Abouhawwash, M.; Askar, S.S.; Sayyouri, M. Octonion-based transform moments for innovative stereo image classification with deep learning. *Complex Intell. Syst.* **2024**, *10*, 3493–3511. [CrossRef]
36. Roos, G. Exceptional symmetric domains. *arXiv* **2008**, arXiv:0801.4076.
37. Luna-Elizarrarás, M.E.; Shapiro, M.; Struppa, D.C.; Vajiac, A. The Bicomplex Numbers. In *Bicomplex Holomorphic Functions. Frontiers in Mathematics*; Birkhäuser: Cham, Switzerland, 2015; ISBN 978-3-319-24866-0. [CrossRef]
38. Cockle, J. On Certain Functions Resembling Qaternions and on a new Imaginary in Algebra. *Lond. Edinb. Dublin Philos. Mag. J. Sci.* **1848**, *33*, 43–59.
39. Cockle, J. On a New Imaginary in Algebra. *Lond. Edinb. Dublin Philos. Mag. J. Sci.* **1849**, *34*, 37–47. [CrossRef]
40. Cockle, J. On the Symbols of Algebra and the Theory of Tessarines. *Lond. Edinb. Dublin Philos. Mag. J. Sci.* **1849**, *34*, 406–410. [CrossRef]
41. Cockle, J. On Impossible Equations, on Impossible Quantities and on Tessarines. *Lond. Edinb. Dublin Philos. Mag. J. Sci.* **1850**, *37*, 281–283. [CrossRef]
42. Segre, C. Le rappresentazioni reali delle forme complesse e gli enti iperalgebrici (Real representations of complex forms and hyperalgebraic bodies). *Math. Ann.* **1892**, *40*, 413–467. [CrossRef]
43. Dragoni, G.S. Sulle funzioni olomorfe di una variabile bicomplessa. *, Reale Acad. D'Italia Mem. Classes Fic. Mat. Nat.* **1934**, *5*, 597–665.
44. Spampinato, N. Estensione nel Campo Bicomplesso di Due Teoremi, del Levi–Civita e del Severi, per le Funzione Olomorfe di Due Variablili Complesse. I, II. *Atti R. Accad. Naz. Lincei Rend* **1935**, *22*, 38–43, 96–102.
45. Rochon, D. Shapiro, M. On Algebraic Properties of Bicomplex and Hyperbolic Numbers. *Analele Univ. Din Oradea. Fasc. Mat.* **2004**, *11*, 71–110.
46. Alfsmann, D. On Families of 2N-dimensional Hypercomplex Algebras Suitable for Digital Signal Processing. In Proceedings of the 14th European Signal Processing Conference, Florence, Italy, 4–8 September 2006; pp. 1–4.
47. Cariow, A.; Cariowa, G. Reduced-Complexity Algorithms for Tessarine Neural Networks. *IEEE Trans. Neural Netw. Learn. Syst.* **2024**, 1–6. [CrossRef] [PubMed]
48. Srivastava, R.K.; Srivastava, N.K. On a class of Entire Bicomplex sequences. *South East. Asian J. Math & Math. Sc.* **2007**, *5*, 47–68.
49. Luna-Elizarrarás, M.E.; Alpay, D.; Struppa, D.C.; Shapiro, M. *Basics of Functional Analysis with Bicomplex Scalars, and Bicomplex Schur Analysis*; Springer: Dordrecht, The Netherlands, 2014. [CrossRef]
50. Torsello, A.; Rodolà, E.; Albarelli, A. Multiview Registration via Graph Diffusion of Dual Quaternions. In Proceedings of the XXIV IEEE Conference on Computer Vision and Pattern Recognition, Colorado Springs, CO, USA, 20–25 June 2011; pp. 2441–2448.
51. Chelnokov, Y.N. Quaternion and Biquaternion Methods and Regular Models of Analytical Mechanics (Review). *Mech. Solids* **2023**, *58*, 2450–2477. [CrossRef]
52. Gai, S.; Huang, X. Regularization method for reduced biquaternion neural network. *Appl. Soft Comput.* **2024**, *166*, 112206. [CrossRef]

53. Pei, S.-C.; Chang, J.-H.; Ding, J.-J. Commutative Reduced Biquaternions and Their Fourier Transform for Signal and Image Processing Applications. *IEEE Trans. Signal Process.* **2004**, *52*, 2012–2031. [CrossRef]
54. Ortolani, F.; Uncini, A. Quaternion Digital Signal Processing: A Hypercomplex Approach to Information Processing. In Proceedings of the 2016 International Siberian Conference on Control and Communications, SIBCON 2016, Moscow, Russia, 12–14 May 2016. [CrossRef]
55. Comminiello, D.; Scarpiniti, M.; Parisi, R.; Uncini, A. Frequency-domain Adaptive Filtering: From Real to Hypercomplex Signal Processing. In Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing, (ICASSP), Brighton, UK, 12–17 May 2019; pp. 7745–7749. [CrossRef]
56. Valkova-Jarvis, Z.; Poulkov, V.; Stoynov, V.; Mihaylova, D. A method for the design of bicomplex orthogonal DSP algorithms for applications in intelligent radio access networks. *Symmetry* **2022**, *14*, 613. [CrossRef]
57. Gou, X.; Liu, Z.; Fan, Z.; Xu, Y. Bicomplex-valued adaptive filtering with application to wind profile prediction. In Proceedings of the 19th International Conference on Digital Signal Processing, Hong Kong, China, 20–23 August 2014; pp. 404–408. [CrossRef]
58. Toyoshima, H.; Higuchi, S. Design of hypercomplex all-pass filters to realize complex transfer functions. In Proceedings of the 2nd International Conference on Information, Communications and Signal Processing, Tokyo, Japan, 9–12 May 1999; No. 2B3.4, pp. 1–5.
59. Schutte, H.-D.; Wenzel, J. Hypercomplex Numbers in Digital Signal Processing. In Proceedings of the IEEE International Symposium on Circuits and Systems, New Orleans, LA, USA, 1–3 May 1990; Volume 2, pp. 1557–1560.
60. Ell, T.A. Quaternion Fourier transforms for analysis of 2-dimensional linear time-invariant partial-differential systems. In Proceedings of the 32nd IEEE Conference on Decision Control, San Antonio, TX, USA, 15–17 December 1993; Volume 2, pp. 1830–1841.
61. Clyde, C.M.; Davenport, M. A commutative hupercomplex algebra with associated function theory. In *Clifford Algebra with Numeric and Symbolic Computations*; Ablamowicz, R., Ed.; Birkhauser: Boston, MA, USA, 1996; pp. 213–227.
62. Berkovich, Y.; Shenkman, A. Introducing Hypernion Numbers for the Analysis of Electric Networks Driven by Nonsinusoidal Sources. *IEEE Power Eng. Rev.* **2001**, *21*, 45–46. [CrossRef]
63. Nos, O.V.; Dudin, A.; Petzoldt, J. The Instantaneous Power Quaternion of the Three-Phase Electric Circuit with Linear Load. In Proceedings of the International Conference of Young Specialists on Micro/Nanotechnologies and Electron Devices, EDM, Erlagol, Russia, 30 June–4 July 2016; pp. 526–531. [CrossRef]
64. Nos, O.V.; Dybko, M.A. Equipment to Improve Energy Efficiency of Power Supply Systems in Mineral Mining Industry. *Gorn. Zhurnal* **2019**, 82–86. [CrossRef]
65. Flifla, J.; Dillenseger, J.L.; Haigron, P.; Coatrieux, J.L. Quaternion Based Registration Method in Medical Imaging. In Proceedings of the 1992 International Biomedical Engineering Days, Istanbul, Turkey, 18–20 August 1992; pp. 161–165.
66. Shu, J.-J.; Ouw, L.S. Pairwise alignment of the DNA sequence using hypercomplex number representation. *Bull. Math. Biol.* **2004**, *66*, 1423–1438. [CrossRef]
67. Shu, J.-J.; Li, Y. Hypercomplex cross-correlation of DNA sequences. *J. Biol. Syst.* **2010**, *18*, 711–725. [CrossRef]
68. Petrovsky, N.; Stankevich, A.; Petrovsky, A. FPSoC using Xilinx Zynq for Medical Image Coding Based on the Quaternionic Paraunitary Filter Banks. In Proceedings of the 17th International Conference on E-Health Networking, Application and Services, HealthCom, Boston, MA, USA, 14–17 October 2015; pp. 596–597. [CrossRef]
69. Lopez, E.; Grassucci, E.; Valleriani, M.; Comminiello, D. Multiview Hypercomplex Learning for Breast Cancer Screening. *arXiv* **2023**, arXiv:2204.05798.
70. Yu, Z.; Li, D.; Du, P.; Luo, W.; Kou, K.I.; Bhatti, U.A.; Benger, W.; Lv, G.; Yuan, L. Hypercomplex Signal Processing in Digital Twin of the Ocean: Theory and application. *IEEE Signal Process. Mag.* **2024**, *41*, 33–48. [CrossRef]
71. Arizaga, J.M.; Cariño, J.; Castañeda, H.; Mercado, D.; Castillo, P. Adaptive Quaternion Control for a Quadcopter Vehicle: Real-Time Validation in Presence of Wind Gusts. In Proceedings of the 2024 International Conference on Unmanned Aircraft Systems (ICUAS), Chania, Greece, 4–7 June 2024; pp. 1261–1266. [CrossRef]
72. Verenik, A.; Parfieniuk, M.; Petrovsky, A. An FPGA Implementation of the Distributed Arithmetic Based Quaternionic Multipliers for Paraunitary Filter Banks. In Proceedings of the 14th International Conference "Mixed Design of Integrated Circuits and Systems", MIXDES, Ciechocinek, Poland, 21–23 June 2007; pp. 605–610. [CrossRef]
73. Shalygin, K.A. Using the Hypercomplex Numbers for Instantaneous Reactive Power Compensation. In Proceedings of the 11th International Conference on Actual Problems of Electronic Instrument Engineering, APEIE, Novosibirsk, Russia, 2–4 October 2012; pp. 168–172. [CrossRef]
74. Nos, O.V.; Nos, A.O. Quaternion-Based Single-Phase Full Bridge VSI Control for Compensation of Voltage Variation and Harmonic Distortion. In Proceedings of the IEEE 25th International Conference of Young Professionals in Electron Devices and Materials (EDM), Altai, Russian Federation, 28 June–2 July 2024; pp. 1080–1086. [CrossRef]
75. Fortuna, L.; Muscato, G.; Xibilia, M.G. A Comparison Between HMLP and HRBF for Attitude Control. *IEEE Trans. Neural Netw.* **2001**, *12*, 318–328. [CrossRef]

76. Zhou, Y.; Zhao, Y.; Yang, Z.; Wang, J. NN Adaptive Control of Spacecraft Attitude. In Proceedings of the 4th World Congress on Intelligent Control and Automation, Shanghai, China, 10–14 June 2002; Volume 3, pp. 2528–2532.
77. Fortuna, L.; Muscato, G.; Xibilia, M.G. An Hypercomplex Neural Network Platform for Robot Positioning. In Proceedings of the IEEE International Symposium on Circuits and Systems (ISCAS '96), 'Connecting the World', Atlanta, GA, USA, 12–15 May 1996; Volume 3, pp. 609–612.
78. Roumeliotis, S.I.; Sukhatme, G.S.; Bekey, G.A. Smoother Based 3D Attitude Estimation for Mobile Robot Localization. In Proceedings of the IEEE International Conference on Robotics and Automation, Detroit, MI, USA, 10–15 May 1999; Volume 3, pp. 1979–1986.
79. Bachmann, E.R.; Duman, I.; Usta, U.Y.; McGhee, R.B.; Yun, X.P.; Zyda, M.J. Orientation Tracking for Humans and Robots Using Inertial Sensors. In Proceedings of the IEEE International Symposium on Computational Intelligence in Robotics and Automation, (CIRA'99), Monterey, CA, USA, 8–9 November 1999; pp. 187–194.
80. Tsiotras, P.; King-Smith, M.; Ticozzi, L. Spacecraft-Mounted Robotics. *Annu. Rev. Control. Robot. Auton. Syst.* **2023**, *6*, 335–362. [CrossRef]
81. Bottlik, I.P. Relationship of nonlinear operations on real and complex signals. *Proc. IEEE* **1982**, *70*, 766–767. [CrossRef]
82. Hahn, L. Multidimensional Complex Signals with Single-Orthant Spectra. *Proc. IEEE* **1992**, *80*, 1287–1300. [CrossRef]
83. Bulow, T.; Sommer, G. Hypercomplex Signals—a Novel Extension of the Analytic Signal to the Multidimensional Case. *IEEE Trans. Signal Process.* **2001**, *49*, 2844–2852. [CrossRef]
84. Martin, K. Complex Signal Processing is Not Complex. *IEEE Trans. Circuits Syst.—I* **2004**, *51*, 1823–1836. [CrossRef]
85. Fré, P.G. Complex and Quaternionic Geometry. In *Advances in Geometry and Lie Algebras from Supergravity*; Theoretical and Mathematical Physics; Springer: Cham, Switzerland, 2018. [CrossRef]
86. Nitta, T.; Kobayashi, M.; Mandic, D.P. Hypercomplex widely linear estimation through the lens of underpinning geometry. *IEEE Trans. Signal Process.* **2019**, *67*, 3985–3994. [CrossRef]
87. Comminiello, D.; Grassucci, E.; Mandic, D.P.; Uncini, A. Demystifying the Hypercomplex: Inductive biases in hypercomplex deep learning. *IEEE Signal Process. Mag.* **2024**, *41*, 59–71. [CrossRef]
88. Valous, N.A.; Hitzer, E.; Vitabile, S. Hypercomplex Signal and Image Processing: Part 1. *IEEE Signal Process. Mag.* **2024**, *41*, 11–13. [CrossRef]
89. Ortolani, F.; Comminiello, D.; Uncini, A. The Widely Linear Block Quaternion Least Mean Square Algorithm for Fast Computation in 3D Audio Systems. In Proceedings of the IEEE International Workshop on Machine Learning for Signal Processing, MLSP-2016, Vietri sul Mare, Italy, 13–16 September 2016. [CrossRef]
90. Panagos, I.I.; Sfikas, G.; Nikou, C. Compressing audio visual speech recognition models with parameterized hypercomplex layers. In Proceedings of the 12th Hellenic Conference on Artificial Intelligence, Corfu, Greece, 7–9 September 2022.
91. Panagos, I.I.; Sfikas, G.; Nikou, C. Visual Speech Recognition using Compact Hypercomplex Neural Networks. *Pattern Recognit. Lett.* **2024**, *186*, 1–7. [CrossRef]
92. Le Bihan, N.; Sangwine, S.J. Quaternion Principal Component Analysis of Color Images. In Proceedings of the International Conference on Image Processing, Barcelona, Spain, 14–17 September 2003; Volume 1, p. I-809-12.
93. An, M.L.; Wang, W.J.; Zhao, Z. Digital Watermarking Algorithm Research of Color Images Based on Quaternion Fourier Transform. In Proceedings of the 8th Symposium on Multispectral Image Processing and Pattern Recognition (MIPPR)—Multispectral Image Acquisition, Processing, and Analysis, Wuhan, China, 26–27 October 2013; Volume 8917. [CrossRef]
94. Li, Y.F. Color Image Restoration by Quaternion Diffusion Based on a Discrete Version of the Topological Derivative. In Proceedings of the 7th International Congress on Image and Signal Processing (CISP), Dalian, China, 14–16 October 2014; pp. 201–206, ISBN 978-1-4799-5835-1.
95. BenYoussef, N.; Bouzid, A. Color Edge Detection using Quaternion Convolution and Vector Gradient. In Proceedings of the 12th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications (VISIGRAPP 2017), Porto, Portugal, 1–27 February 2017; Volume 4, pp. 135–139. [CrossRef]
96. Grassucci, E.; Sigillo, L.; Uncini, A.; Comminiello, D. Hypercomplex Image-to-Image Translation. In Proceedings of the 2022 International Joint Conference on Neural Networks (IJCNN), Padua, Italy, 18–23 July 2022; pp. 1–8.
97. Dizon, N.D.; Hogan, J.A. Holistic Processing of Color Images Using Novel Quaternion-Valued Wavelets on the Plane: A Promising Transformative Tool. *IEEE Signal Process. Mag.* **2024**, *41*, 51–63. [CrossRef]
98. Bardos, A.J.; Sangwine, S.J. Measuring Noise in Colour Images. In Proceedings of the IEEE Colloquium on Non-Linear Signal and Image Processing (Ref. No. 1998/284), London, UK, 22 May 1998; pp. 8/1–8/4.
99. Tao, J.W. Performance Analysis for Interference and Noise Canceller Based on Hypercomplex and Spatio-Temporal-Polarisation Processes. *IET Radar Sonar Navig.* **2013**, *7*, 277–286; ISSN 1751-8784, eISSN 1751-8792. [CrossRef]
100. Mei, X.; Lin, B.; Li, S. Adaptive Equalization Algorithm Based on Quaternion in Pulsed Noise Environment. In Proceedings of the 2024 IEEE/CIC International Conference on Communications in China (ICCC), Hangzhou, China, 7–9 August 2024; pp. 439–444. [CrossRef]

101. Jones, C., III; Abbott, A.L. Color Face Recognition by Hypercomplex Gabor Analysis. In Proceedings of the 7th International Conference on Automatic Face and Gesture Recognition, (FGR 2006), Southampton, UK, 10–12 April 2006; pp. 126–131. [CrossRef]
102. Wong, W.K.; Lee, G.C.; Loo, C.K.; Lock, R. Quaternion Based Fuzzy Neural Network Classifier for Mpik Dataset's View-Invariant Color Face Image Recognition. *Informatica* **2013**, *37*, 181–192.
103. Carre, P.; Denis, P.; Fernandez-Maloigne, C. Spatial Color Image Processing Using Clifford Algebras: Application to Color Active Contour. *Signal Image Video Process.* **2014**, *8*, 1357–1372. [CrossRef]
104. Vaijayanthi, S.; Arunnehru, J. Facial Expression Recognition Using Hyper-Complex Wavelet Scattering and Machine Learning Techniques. In *Proceedings of the 6th International Conference on Advance Computing and Intelligent Engineering; Lecture Notes in Networks and Systems*; Pati, B., Panigrahi, C.R., Mohapatra, P., Li, K.C., Eds.; Springer: Singapore, 2023; Volume 428. [CrossRef]
105. Lopez, E.; Uncini, A.; Comminiello, D. Hierarchical Hypercomplex Network for Multimodal Emotion Recognition. In Proceedings of the IEEE 34th International Workshop on Machine Learning for Signal Processing (MLSP), London, UK, 22–25 September 2024. [CrossRef]
106. Ortolani, F.; Comminiello, D.; Scarpiniti, M.; Uncini, A. On 4-Dimensional Hypercomplex Algebras in Adaptive Signal Processing. In *Neural Advances in Processing Nonlinear Dynamic Signals*; WIRN 2017, Smart Innovation, Systems and Technologies; Esposito, A., Faundez-Zanuy, M., Morabito, F., Pasero, E., Eds.; Springer: Berlin/Heidelberg, Germany, 2019; Volume 102. [CrossRef]
107. Valle, M.E.; Lobo, R.A. Hypercomplex-valued Recurrent Correlation Neural Networks. *Neurocomputing* **2021**, *432*, 111–123. [CrossRef]
108. Kycia, R.; Niemczynowicz, A. Hypercomplex Neural Network in Time Series Forecasting of Stock Data. *arXiv* **2024**, arXiv:2401.04632.
109. Lopez, E.; Grassucci, E.; Capriotti, D.; Comminiello, D. Towards Explaining Hypercomplex Neural Networks. *arXiv* **2024**, arXiv:2403.17929.
110. Hazubski, S.; Hoppe, H. Self-organizing Hypercomplex-valued Adaptive Network. *Neurocomputing* **2024**, *607*, 128429. [CrossRef]
111. Selvakumaran, T.V.; Sundar Rajan, B. Block-Coded Modulation Using Two-Level Group Codes Over Generalized Quaternion Groups. *IEEE Trans. Inf. Theory* **1999**, *45*, 365–372. [CrossRef]
112. Sethuraman, B.A.; Rajan, B.S. An Algebraic Description of Orthogonal Designs and The Uniqueness of the Alamouti Code. In Proceedings of the IEEE Global Telecommunications Conference, (GLOBECOM'02), Taipei, Taiwan, 17–21 November 2002; Volume 2, pp. 1088–1092.
113. Nagase, T.; Komata, M.; Araki, T. Secure Signals Transmission Based on Quaternion Encryption Scheme. In Proceedings of the 18th International Conference on Advanced Information Networking and Applications, (AINA 2004), Fukuoka, Japan, 29–31 March 2004; Volume 2, pp. 35–38.
114. Pöppelbaum, J.; Schwung, A. Time Series Compression using Quaternion Valued Neural Networks and Quaternion Backpropagation. *arXiv* **2024**, arXiv:2403.11722.
115. Dukhnich, E.; Chefranov, A. Hypercomplex Numbers in Cryptosystems. In *Current Problems of Applied Mathematics and Computer Systems*; CPAMCS 2023. Lecture Notes in Networks and Systems; Alikhanov, A., Tchernykh, A., Babenko, M., Samoylenko, I., Eds.; Springer: Cham, Switzerland, 2023; Volume 1044. [CrossRef]
116. Urbanovichab, P.P.; Shutkob, N.P. Usage of Hypercomplex Numbers in a Cryptographic Key Agreement Protocol Based on Neural Networks. *J. Belarusian State Univ. Math. Inform.* **2024**, *2*, 81–92.
117. Marins, J.L.; Yun, X.; Bachmann, E.R.; McGhee, R.B.; Zyda, M.J. An Extended Kalman Filter for Quaterni-on-Based Orientation Estimation Using MARG Sensors. In Proceedings of the IEEE/RSJ International Conference on Proceedings Intelligent Robots and Systems, Maui, HI, USA, 29 October–3 November 2001; Volume 4, pp. 2003–2011.
118. Redfield, S.A.; Huynh, Q.Q. Hypercomplex Fourier Transforms Applied to Detection for Side-Scan Sonar. In Proceedings of the Oceans '02 MTS/IEEE, Biloxi, MI, USA, 29–31 October 2002; Volume 4, pp. 2219–2224.
119. Akay, O.; Erözden, E. Employing Fractional Autocorrelation for Fast Detection and Sweep Rate Estimation of Pulse Compression Radar Waveforms. *Signal Process.* **2009**, *89*, 2479–2489. [CrossRef]
120. Shuai, S.; Liu, A.; Yu, C.; Zhe, L.; Zhao, Q. HFSWR Polarization DOA Estimation Based on Quaternion. In *Wireless and Satellite Systems*; WiSATS 2021. Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering; Guo, Q., Meng, W., Jia, M., Wang, X., Eds.; Springer: Cham, Switzerland, 2022; Volume 410. [CrossRef]
121. Wang, X.; Feng, Y.; Lv, X.; Wang, R. Geometric Algebra based 2D-DOA Estimation for Non-circular Signals with an Electromagnetic Vector Array. *Digit. Signal Process.* **2024**, *148*, 104459. [CrossRef]
122. Wong, M.; Eagleson, R. Canonical Decomposition of Affine Motion for Visual Servoing. In Proceedings of the IEEE International Conference on Systems, Man, and Cybernetics, 'Computational Cybernetics and Simulation', Orlando, FL, USA, 12–15 October 1997; Volume 1, pp. 767–772.
123. Yamamoto, H.; Aoshima, N. Three Dimensional Signal Processing Based on Quaternion. In Proceedings of the 41st SICE Annual Conference (SICE 2002), Osaka, Japan, 5–7 August 2002; Volume 3, pp. 1420–1424.

124. Yun, X.; Lizarraga, M.; Bachmann, E.R.; McGhee, R.B. An improved Quaternion-Based Kalman Filter For Real-Time Tracking of Rigid Body Orientation. In Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems, (IROS 2003), Las Vegas, NV, USA, 27–31 October 2003; Volume 2, pp. 1074–1079.
125. Vieira, G.; Grassucci, E.; Valle, M.E.; Comminiello, D. Dual Quaternion Rotational and Translational Equivariance in 3D Rigid Motion Modelling. In Proceedings of the IEEE International Workshop on Machine Learning for Signal Processing (MLSP), Rome, Italy, 17–20 September 2023.
126. Bojesomo, A.; Liatsis, P.; Al Marzouqi, H. Deep Hypercomplex Networks for Spatiotemporal Data Processing: Parameter efficiency and superior performance. *IEEE Signal Process. Mag.* **2024**, *41*, 101–112. [CrossRef]
127. Cadman, L.; Tjahjadi, T. Simultaneous Feature Tracking and Three-Dimensional Object Reconstruction from an Image Sequence. In Proceedings of the International Conference on Image Processing, Thessaloniki, Greece, 7–10 October 2001; Volume 2, pp. 391–394.
128. Cai Xingyu; Zhou Wujie; Cen Gang; Qiu Weiwei, Saliency Detection for Stereoscopic 3D Images in the Quaternion Frequency Domain. *3D Res.* **2018**, *9*, 22. [CrossRef]
129. He, X.; Chen, Y.; Zhang, L. Hyper-Complex Number Quaternion System-Based Method for Image Representation. *Signal Image Video Process.* **2024**, *18*, 7899–7907. [CrossRef]
130. Condurache, D. Advances Representation of Higher-Order Kinematics of Motion. Hypercomplex Lie Groups and Lie Algebras. In *25th International Symposium on Measurements and Control in Robotics, ISMCR 2023*; Doroftei, I., Kiss, B., Baudoin, Y., Taqvi, Z., Keller Fuchter, S., Eds.; Mechanisms and Machine Science; Springer: Cham, Switzerland, 2023; Volume 154. [CrossRef]